

Object Detection Using Deep Learning Algorithm

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

This project addresses the imperative of labor safety by employing deep learning algorithms for object detection in industrial environments. Focusing on the You Only Look Once (YOLO) algorithm, our approach combines the efficiency of real-time object detection with the urgency of laboring scenarios. The YOLO algorithm, known for its speed and accuracy, is utilized to identify and classify objects, ensuring prompt recognition of potential hazards in the workplace.

Object detection is a fundamental task in computer vision with wide-ranging applications in surveillance, autonomous vehicles, robotics, and smart systems. This project presents an efficient and real-time object detection approach using the YOLO (You Only Look Once) algorithm. Unlike traditional detection methods that apply region proposals followed by classification, YOLO treats object detection as a single regression problem, directly predicting bounding boxes and class probabilities from full images in one evaluation.

Among the various object detection algorithms developed in recent years, **YOLO (You Only Look Once)** has gained significant attention due to its exceptional speed and accuracy. Unlike traditional object detection techniques that utilize region proposal methods followed by classification (e.g., R-CNN and Fast R-CNN), YOLO approaches the task as a single regression problem, enabling real-time detection by predicting bounding boxes and class probabilities directly from entire images in a single forward pass of a convolutional neural network

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CHAPTER 1

INTRODUCTION

1.1 Introduction to Object Detection:

Object detection is a fundamental task in computer vision that involves identifying and localizing objects within an image or video frame. Unlike image classification, which determines the presence of objects in an entire image, object detection goes further by precisely locating and classifying multiple objects simultaneously. This capability has a wide range of applications across various industries, including surveillance, autonomous driving, medical imaging, and retail.

Traditionally, object detection relied on handcrafted features and classical machine learning algorithms, such as Viola-Jones and Histogram of Oriented Gradients (HOG). However, with the advent of deep learning, particularly convolutional neural networks (CNNs), object detection has seen significant advancements in accuracy and efficiency.

Deep learning-based object detection methods leverage CNNs to automatically learn discriminative features from raw pixel data. These methods typically follow one of two main approaches: region-based methods and single-shot methods.

Region-based methods, such as Faster R-CNN and Mask R-CNN, first generate region proposals that are likely to contain objects. These proposals are then classified and refined to produce the final object detections. While region-based methods achieve high accuracy, they are computationally intensive and may suffer from slower inference speeds.

Single-shot methods, exemplified by models like YOLO (You Only Look Once) and SSD (Single Shot Multibox Detector), take a different approach by directly predicting object bounding boxes and class probabilities from a single pass through the network. These methods are faster and more suitable for real-time applications but may sacrifice some accuracy compared to region-based

In recent years, the field of object detection has seen continuous

advancements, with new architectures, training techniques, and evaluation metrics being developed. The ongoing research aims to improve both the accuracy and efficiency of object detection systems, making them more robust and applicable to a wider range of scenarios.

In summary, object detection plays a crucial role in computer vision, enabling machines to understand and interact with the visual world. With the rapid progress in deep learning and the proliferation of data-driven approaches, the capabilities of object detection systems continue to evolve, opening up new opportunities for innovation and application.

The development of the YOLO (You Only Look Once) algorithm has completely changed the field of object detection in photos and videos in terms of efficiency and accuracy. In order to develop a reliable system for accurate labelling and real-time object identification in a variety of scenarios, this study explores and applies You only look once(YOLO). By partitioning an image into a grid and concurrently predicting bounding boxes and class probabilities, You only look once(YOLO) 's innovative methodology makes it possible to quickly and accurately identify several items within complicated scenarios.

This project's main goal is to address the fundamental problems with computer vision, namely with localization, classification, and object recognition. Across a wide range of industries, including but not limited to autonomous vehicles, retail, healthcare, and surveillance, object detection is still a crucial duty. Identifying items in a variety of situations quickly and reliably is essential for improving decision-making, task automation, and maintaining efficiency and safety in many different fields.

The first step in the project is thorough data curation and annotation, where a variety of datasets are carefully prepared and enhanced with masks or bounding boxes that define objects of interest. The You only look once(YOLO) model is trained on this annotated data base by utilizing transfer learning from large-scale datasets such as or ImageNet. The system strives to maximize object detection speed and accuracy by utilizing pre-trained models and fine-tuning parameters, meeting certain use cases and scenarios.

The project also emphasizes how important user-friendly integration and deployment are. It is essential to have an easy-to-use interface that can use the trained You only look once(YOLO) model for accurate labelling and real-time item recognition in pictures or video streams. The adaptability of this integration must be guaranteed, supporting a range of output formats and allowing for smooth system or application interfaces.

This project has a broad range of possible effects. When You only look once(YOLO) for object detection and labelling is successfully implemented, computer vision applications are expected to advance, giving industries access to greater safety measures, automated processes, and increased operational efficiency. The objective of this project is to develop a flexible and dependable framework that can handle the intricate requirements of contemporary object recognition and classification jobs, hence contributing to the advancement of object detection systems.

1.2 Evolution of Deep Learning in Object Detection:

The evolution of deep learning in object detection has been a transformative journey, driven by continual innovation and research efforts. Initially, the field witnessed the rise of region-based methods such as R-CNN, which leveraged CNNs for object localization and classification. However, these methods were computationally expensive due to the need for region proposal generation and subsequent classification. The introduction of single-shot methods like YOLO and SSD revolutionized object detection by enabling real-time processing through direct prediction of object bounding boxes and

class probabilities. Building upon these foundations, subsequent advancements focused on refining detection accuracy, speed, and versatility. Anchor-based methods, as seen in Faster R-CNN and SSD, improved localization accuracy by predicting object positions relative to predefined anchor boxes. Conversely, anchor-free approaches like CenterNet and FCOS simplified the detection process by directly predicting object centroids or bounding box offsets without anchor boxes. Hybrid models like YOLOv4 and EfficientDet combined the strengths of single-shot and two-stage techniques to achieve state-of-the-art performance across various datasets and scenarios.

Furthermore, attention mechanisms and contextual information integration have enhanced the ability of models to capture intricate details and contextual dependencies, leading to further improvements in detection accuracy and robustness. As a result of these advancements, modern object detection systems have become indispensable in a myriad of applications, ranging from surveillance and security to autonomous vehicles and medical imaging, driving innovation and progress in computer vision research and industry applications.

1.3 Introduction of YOLOv8:

Object detection is a critical task in computer vision, enabling machines to identify and locate objects within images or video frames. YOLO, short for You Only Look Once, is a groundbreaking object detection algorithm that has significantly advanced the field by providing a fast and accurate solution. YOLO introduces a unique approach by framing object detection as a single regression problem, where the model predicts bounding boxes and class probabilities directly from the entire image in one pass through the network.

YOLOv8 represents the latest iteration in the YOLO series, incorporating advancements in deep learning methodologies and architectural enhancements to further improve accuracy and efficiency. Building upon its predecessors, YOLOv8 leverages state-of-the-art techniques such as multi-scale feature fusion, anchor-based prediction, and backbone network optimization to achieve superior performance in object detection tasks.

The key innovation of YOLO and its subsequent versions lies in their ability to achieve real-time object detection while maintaining high accuracy. By processing the entire image at once, YOLO eliminates the need for complex region proposal mechanisms, leading to faster inference speeds and reduced computational overhead. This makes YOLO and its variants particularly well-suited for applications requiring rapid and efficient object detection, such as surveillance, autonomous driving, and augmented reality.

YOLOv8 builds upon the success of its predecessors by introducing refinements and optimizations aimed at further enhancing detection performance. With its state-of-the-art architecture and efficient implementation, YOLOv8 continues to push the boundaries of object detection, opening up new possibilities for real-world applications and advancing the field of computer vision.

In the realm of computer vision, object detection serves as a fundamental building block for various applications, from autonomous vehicles to security surveillance systems. YOLO, standing for You Only Look Once, emerged as a revolutionary approach to object detection, fundamentally altering the landscape of this field. YOLO fundamentally reconceptualized object detection as a single unified regression problem, setting it apart from traditional methods that relied on multiple stages and complex region proposal mechanisms.

Aim and objective:

The goal is to precisely apply You only look once(YOLO) for real-time object detection. This includes setting up the algorithm, curating the dataset with thorough annotations, and training the model to increase accuracy. Repetitive enhancements are guided by metrics like mean average precision. Practical usability across a range of environments is ensured through interface development, scalability checks, and deployment in real-time scenarios. Extensive documentation reveals the strengths and weaknesses of the system by covering approaches, problems, and solutions. The goal is to create a reliable object detection system that can quickly identify and classify objects in a variety of domains. This accomplishment increases its application potential and boosts safety and efficiency in domains such as robotics, autonomous vehicles, and surveillance.

The study of YOLOv8 encompasses a multifaceted approach aimed at thoroughly investigating the capabilities and potential advancements of this object detection model. Firstly, it delves into the architectural enhancements introduced in YOLOv8 compared to its predecessors, such as YOLOv7, to discern their contributions to improved performance, accuracy, and efficiency in object detection tasks.

Rigorous performance evaluation experiments are conducted to assess YOLOv8's detection accuracy, precision, recall, and speed across diverse datasets and scenarios, facilitating comparisons with other state-of-the-art models. Additionally, the study explores various training strategies and optimization techniques employed in training YOLOv8, including data augmentation, transfer learning, and hyperparameter tuning, to analyze their impact on model convergence, generalization, and robustness.

CHAPTER 2

LITERATURE SURVEY

1. The paper by Bruckner and Daniel published on 2014. The report likely focuses on a visualization tool or system designed to aid in the diagnostic process of deep machine learning pipelines. The purpose of such a system would likely be to provide insights into the inner workings of complex machine learning models, allowing researchers and practitioners to better understand and debug their systems [1].
2. It was published by John Wiley & Sons in 2018. The book likely delves into various aspects related to the Internet of Things (IoT), including technical challenges, practical applications, and ethical considerations. Given the interdisciplinary nature of IoT, it may cover topics such as connectivity, security, privacy, data management, and societal impacts of IoT technologies[2].
3. Proposal likely explores strategies and procedures to upgrade protest discovery calculations by leveraging semantic connections between objects in pictures. This might include utilizing relevant data or conditions between objects to progress the precision and strength of question discovery frameworks [3].
4. The paper examines the usage of cleverly observation as an edge arrange benefit, transitioning from conventional strategies such as Haar cascades and Support Vector Machines (SVM) to a more lightweight Convolutional Neural Network (CNN) approach. The center is likely on moving forward the effectiveness and exactness of observation frameworks utilizing edge computing technologies[4].
5. In their study, Liu, W., Anguelov The paper likely introduces the SSD (Single Shot Multi Box Detector) framework, which is a real-time object detection method that operates directly on input images and predicts bounding boxes and class probabilities simultaneously. This approach is efficient and effective for various object detection tasks[5].
6. Ren, S., He, K., Girshick et al The paper introduces the Faster R-CNN

framework, which aims to improve the speed and accuracy of object detection using region proposal networks (RPNs). Faster R-CNN integrates the region proposal step with the object detection step, resulting in a more efficient and effective object detection system compared to previous methods. This work has had significant impact in the field of computer vision and has been widely adopted for various object detection applications[6].

7. Lin, T. Y., Dollár et al. The paper introduces Feature Pyramid Networks (FPN), a framework designed to improve object detection by addressing the scale variation problem. FPN utilizes a top-down architecture with lateral connections to build high-level semantic feature maps at multiple scales. This enables more accurate detection of objects at different sizes and resolutions, leading to better performance on object detection tasks[7].
8. It was published by K., Gkioxari This paper provides The paper introduces Mask R-CNN, an extension of the Faster R-CNN framework, which adds a branch for predicting segmentation masks alongside the existing branches for object detection and bounding box regression. Mask R-CNN achieves state-of-the-art performance in instance segmentation tasks by enabling pixel- level segmentation of objects in images. This paper has significantly advanced the field of computer vision, particularly in the area of instance segmentation.[8].

CHAPTER-3

PROPOSED METHODOLOGY

3.1 Introduction:

Compared to other methods yolo can predict an output with more accuracy and speed. There are certain limitations for other methods for sensor based method high cost, difficult to differentiate the type of object. For parameter variation method it is limited to certain power applications. For sensing pattern-based it cannot detect small ferromagnetic materials accurately.

YOLO, or You Only Look Once, revolutionized object detection by presenting a unified approach that processes the whole picture at once instead of in different stages. This imaginative strategy significantly diminishes computational complexity and deduction time. YOLO works by isolating the input picture into a network and foreseeing bounding boxes and course probabilities for each framework cell at the same time. Not at all like conventional strategies that create region proposals and after that classify them, YOLO specifically predicts the bounding boxes in conjunction with their related course probabilities in a single forward pass through the neural arrange.

Speed: As YOLO anticipate objects in real-time, it moves forward location speed. Compared to other calculations yolo can perform much speedier running at 45 outlines per moment. Another fundamental distinction is YOLO has the capability to see total pictures at as it were once which isn't show in previous methods. We are going run the picture on CNN for as it were one time at run time. All the testing and preparing parameters are as same as between fast Yolo and YOLO.

Tall precision: YOLO features a high prediction capacity that gives the leading comes about with less foundation botches. There are a few distinctive heuristics to extend yolo exactnesses like cosine learning rate scheduler, information increase, bunch normalization (synchronized) and picture mix-up

Learning capabilities: yolo features a tall learning capability, which permits one to discover out the designs of the objects and apply them within the handle of discovery of objects. Yolo obtained the question discovery by division of an picture into N lattices, of rise to measurements $S \times S$. Based on the COCO dataset (common objects in setting), this calculation can distinguish classes of 80 COCO objects: transport, individual, car, Bike, motorbike, plane, truck, prepare, vessel. High-resolution

classifier: For the most part, some time recently training, a genuine YOLO neural arrange employments 224×224 pixels and at that point changes to 448×448 P whereas acknowledgment. Within the handle of changing from one show to another show which is classification to discovery demonstrate, the demonstrate alters to the classification of the picture. YOLO adaptation 2 segments the earlier preparing strategy into fair two steps:

prepare the network with 224×224 Pixels and change over the pixels to 448×448 . Physically based preparing: For the location of pictures with diverse resolutions at that point this demonstrate empowers the same some time recently the organize. We know the preparing speed is quick when the given input is little and when the given input measure is tall then the speed of preparing is moo. Physically based preparing can too progress its precision so that there will be a great adjust between speed and accuracy

3.2 Flowchart:

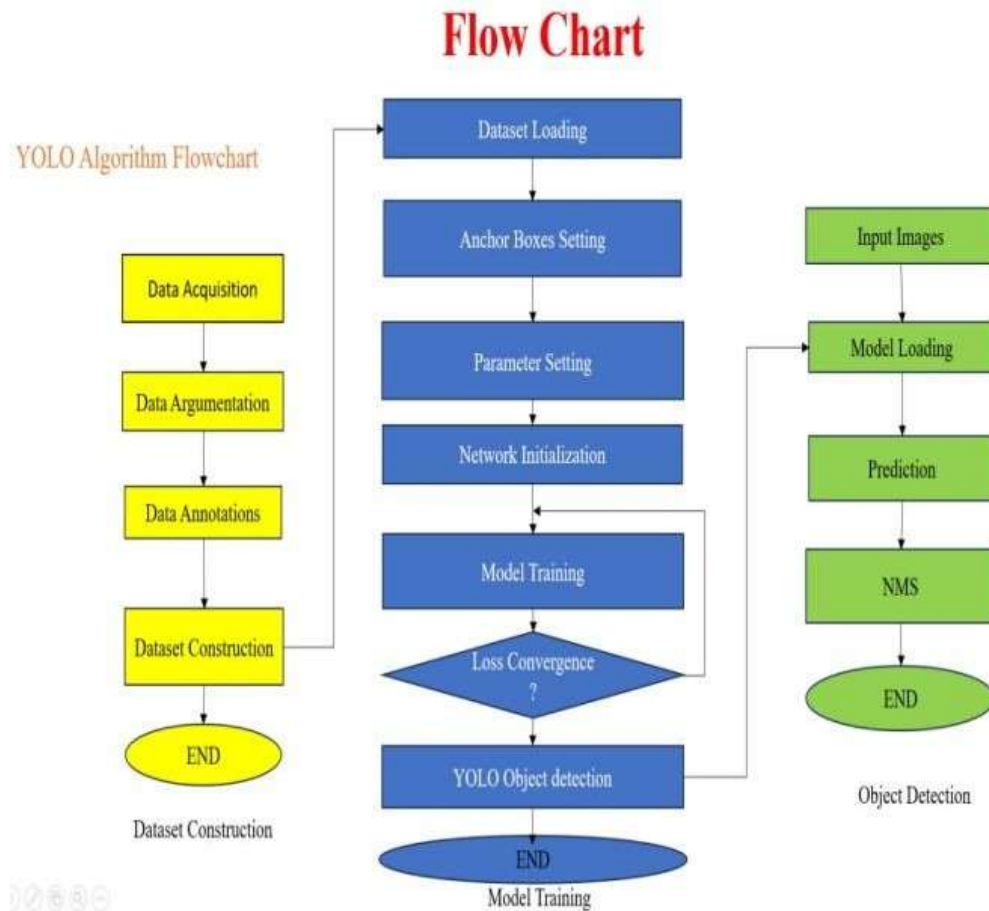


Fig3.2 Flowchart of Data Set Collection, Training, Object Detection

3.3 Custom Dataset Collection:

Collection of custom dataset from nearby charging station and collecting different images for different objects (ex: helmet, tray, bucket and box). With more number of images the accuracy of the detection will be increased. Quality identification of the images will give more accuracy. A well-built dataset increases generalization, reduces biases, and enables the model to correctly identify objects it was trained on, enhancing the system's overall performance and flexibility.

- Data Acquisition
- Data Argumentation
- Data Annotation
- Dataset construction

Data Acquisition :

The process of sampling signals that gauge actual physical circumstances and transforming the resulting samples into digital numeric values that a computer can manipulate is known as data acquisition.

It is generally acknowledged that data acquisition is separate from prior methods of recording to tape recorders or paper charts. In contrast to those techniques, the signals are transformed from the analog to the digital domain before being stored on a digital medium like flash memory, or hard disk drives.

Data Argumentation:

1. Data augmentation uses existing data to create modified copies of datasets, which are then used to artificially increase the training set. It involves making little adjustments to the dataset or creating new data points using deep learning.
2. It is used to avoid models being overfit.
3. It is used when the first training set is not big enough.
4. It is used to increase the model's precision.
5. It is used to cut down on the costs associated with labeling and cleaning the raw dataset.

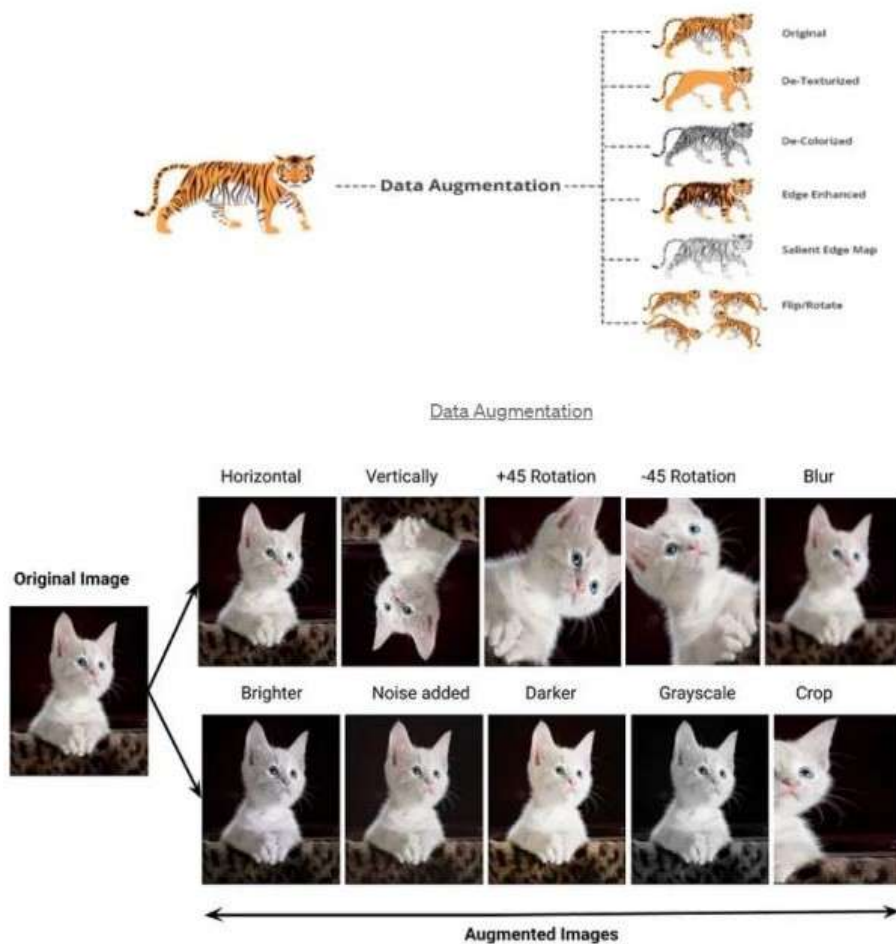


Fig3.3 augmented images

Data Annotation:

The process of labeling data with pertinent tags to make it simpler for computers to understand and analyze is known as data annotation. Data annotators are required to classify this data as precisely as they can, whether it takes the form of photos, text, audio, or video.

Dataset construction:

A set of data used to train the model is known as a deep learning dataset. To educate the deep learning algorithm how to make predictions, a dataset is used as an example. Text data is one of the common sorts of data. image information.

Data Argumentation Code:

```
train_path,
classes=[image_folder],
class_mode =None,
color_mode=image_color_mode,
target_size=target_size,
batch_size=batch_size,
save_to_dir=save_to_dir,
save_prefix=image_save_prefix,
seed=seed)
train_generator=zip(image_generator)
for (img) in train_generator:
    yield (img)
#Data Generation for Augmentation
data_gen_args=dict(rotation_range=.5,
                    width_shift_range=0.05,
                    height_shift_range=0.05,
                    shear_range=0.05,
                    zoom_range=0.05,
                    horizontal_flip=True,
                    fill_mode='nearest')
myGenerator = trainGenerator(15,'Data','image',data_gen_args,save_to_dir = "Data/aug" )
from keras.preprocessing.image import ImageDataGenerator
# Processing
import numpy as np
import os
import glob
```

```

import numpy as np
import os
import glob
import skimage.io as io
import skimage.transform as trans
num_batch = 6
for i ,batch in enumerate(myGenerator):
    if(i >= num_batch):
        break
    #storing data
import matplotlib.pyplot as plt
import random
image_arr=geneTrainNpy("Data/aug/")

```

Giving input data to the code:

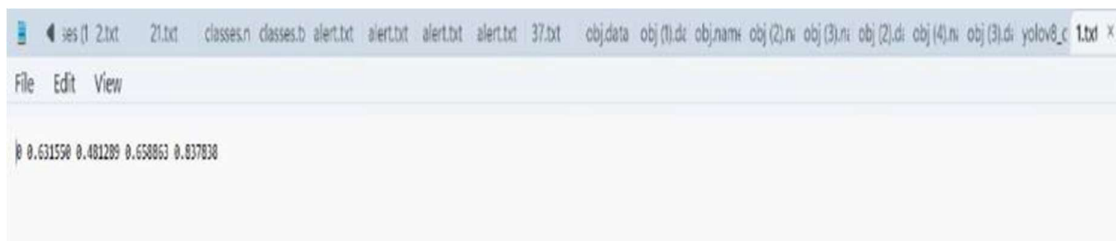
- 1.First collect different image data set
- 2.Later on create a folder called NEW PROJECT
- 3.Create a folder called Data in the NEW PROJECT
- 4.Create a two sub folders called Image and Aug
- 5.Store the collected images in the Image and Aug folders separately
- 6.Convert the photos in the Data folder to the zip format
- 7.Then later on open google collab and open the code
- 8.Give the collected dataset as the input
- 9.Later when we compile the code
 10. We can see the output will be visible in the aug folder
 - 11.According to the input we given

3.4 Labelling:

You only look once(YOLO) labelling entails class labels being assigned to the bounding boxes that surround objects of interest in the images. Every bounding box indicates the position of an object, which is specified by its dimensions (width, height) and coordinates (x, y). To guarantee that the model recognizes and categorizes objects correctly during training and inference, accurate labelling is essential. You only look once(YOLO) can effectively learn object features when it is provided with accurate and consistent annotations across a wide range of datasets. The ability of the model to identify different objects in pictures or videos is improved by properly labelled data, which also greatly increases the model's accuracy and robustness in real-time object detection tasks.

Annotation:

After converting Data Argumentation file into the text file we got the inference as (object class ,height ,width ,x , y axis).



Process for You only look once(YOLO):

1. Create a folder called data
2. Give all the input dataset to that folder
3. Create a folder called output_images
4. The results will be saved to that folder after execution of the code

5. Create a folder called test images
6. Then give testing images to this folder
7. Then give testing images to this folder
8. Create a folder called training results then the results will be saved this folder

3.5 You only look once(YOLO) Code:

1. This command will try to download and install the "ultralytics" package and its dependencies from PyPI when run in a Python environment, enabling the use of its features in Python code.



```
# Pip install (recommended)
!pip install ultralytics
... Looking in indexes: https://pypi.org/simple, https://us-python.pkg.dev/colab-wheels/public/simple/
```

2.import os:

Module, allowing interaction with the operating system, including file and directory manipulation.

import shutil:

used for copying, and deleting the files

import random:

which is used for generating random numbers and performing random selections.

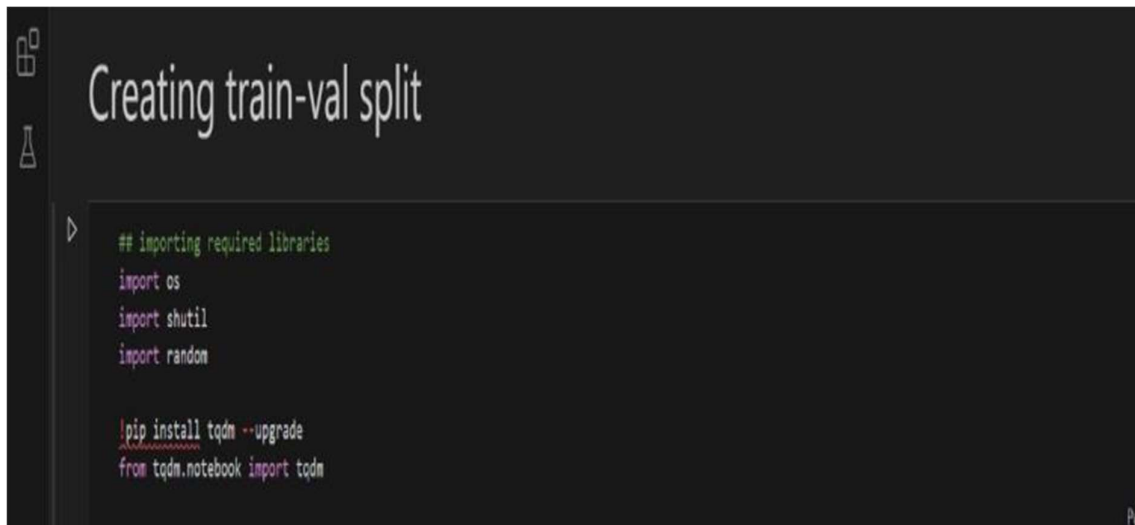
!pip install tqdm –upgrade:

uses the! in a collaborative notebook or other similar environment to carry out a shell command. Using the Python Package Index (PyPI), this command installs or upgrades the tqdm package using pip. If tqdm is already installed,

the `--upgrade` flag guarantees that it will be updated to the most recent version.

from tqdm.notebook import tqdm:

A specific module named `tqdm` is imported from the `tqdm` package with this line. In this instance, the `tqdm` module is imported from the `tqdm notebook` subpackage. In Python scripts or notebooks, loops and tasks can be tracked using the progress bar provided by the `tqdm` library.



```
## importing required libraries
import os
import shutil
import random

!pip install tqdm --upgrade
from tqdm.notebook import tqdm
```

3.Importing from Drive to collab:

Connecting google drive to collab and importing the drive folder to colla

```
## connecting to the google drive

from google.colab import drive
drive.mount('/content/drive')
```

4. Creating path files for images

```
train_path_img = "./yolo_data/images/train/"  
train_path_label = "./yolo_data/labels/train/"  
val_path_img = "./yolo_data/images/val/"  
val_path_label = "./yolo_data/labels/val/"  
test_path = "./yolo_data/test"
```

train_path_img = "./yolo_data/images/train/":

Specifies a variable name that corresponds to the directory path containing the YOLO dataset's training images.

Gives the location of the training image directory. The current directory is indicated by the ./ . There is an images subfolder and a train subfolder in the yolo_data folder, which contain training images.

Similarly it will create folder label, validation, testing in the same way from

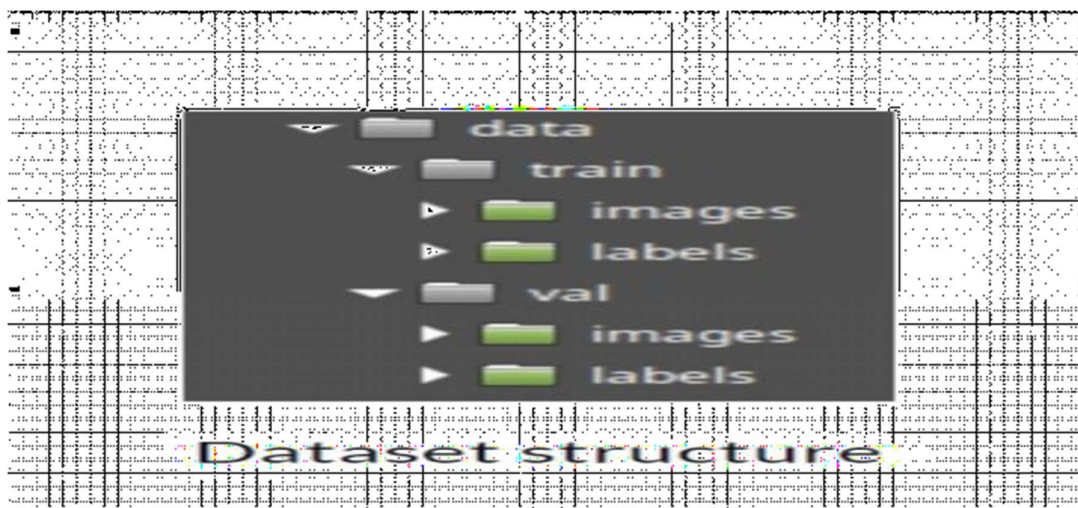


Fig 3a Dataset Structure

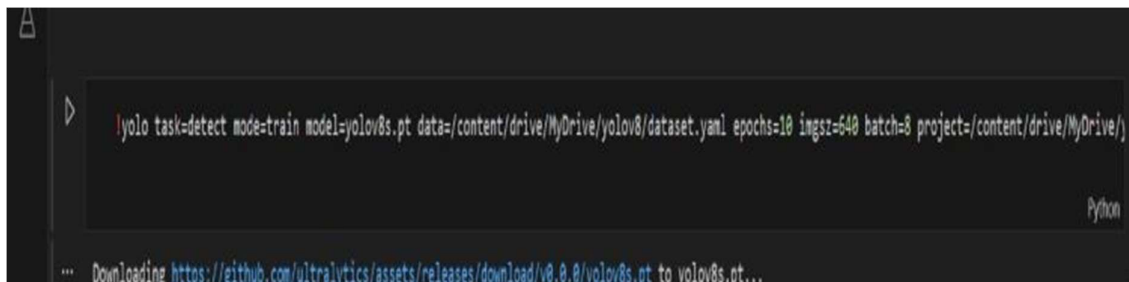
There will be another folder in the train that is test or predict images

5. Dividing into 80:20

Splitting the data into testing and training in 20percent to 80 percent ratio using python code in google colabartory

6. Training:

Training the data using this command

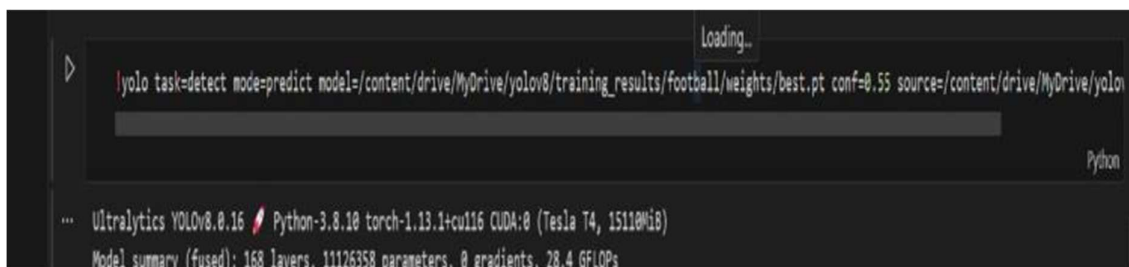
A terminal window with a dark background. The command entered is: `!yolo task=detect mode=train model=yolov8.pt data=/content/drive/MyDrive/yolov8/dataset.yaml epochs=10 imgsz=640 batch=8 project=/content/drive/MyDrive/`. The terminal shows a progress bar and a message: `... Downloading https://github.com/ultralytics/assets/releases/download/v0.0.0/yolov8s.pt to yolov8s.pt...`. The word "Python" is visible in the bottom right corner of the terminal window.

```
!yolo task=detect mode=train model=yolov8.pt data=/content/drive/MyDrive/yolov8/dataset.yaml epochs=10 imgsz=640 batch=8 project=/content/drive/MyDrive/
```

... Downloading <https://github.com/ultralytics/assets/releases/download/v0.0.0/yolov8s.pt> to yolov8s.pt...

Epochs is used to train the data number of times and images 640, batch 8 is inbuilt commands in YOLOV8

7. Infrencing:

A terminal window with a dark background. The command entered is: `!yolo task=detect mode=predict model=/content/drive/MyDrive/yolov8/training_results/football/weights/best.pt conf=0.55 source=/content/drive/MyDrive/yolo`. A "Loading..." status is shown above the command. Below the command, a progress bar is visible. At the bottom, the terminal displays: `... Ultralytics YOLOv8.0.16 Python-3.8.10 torch-1.13.1+cu116 CUDA:0 (Tesla T4, 15110MiB) Model summary (fused): 168 layers, 11126358 parameters, 0 gradients, 28.4 GFLOPs`. The word "Python" is visible in the bottom right corner of the terminal window.

```
!yolo task=detect mode=predict model=/content/drive/MyDrive/yolov8/training_results/football/weights/best.pt conf=0.55 source=/content/drive/MyDrive/yolo
```

... Ultralytics YOLOv8.0.16 Python-3.8.10 torch-1.13.1+cu116 CUDA:0 (Tesla T4, 15110MiB)
Model summary (fused): 168 layers, 11126358 parameters, 0 gradients, 28.4 GFLOPs

Predict is used to predict and collecting the data from drive in yolov8 folder in that training results folder and (class name), weights and best.ptconf=0.55: Establishes the object detection confidence threshold. Objects that are detected and have a confidence level greater than 0.55 (55%) are probably legitimate detections.

Train_images command is used to store the results in that folder

```
!cp -r /content/runs/detect/predict /content/drive/MyDrive/yolov8/output
```

The output will be stored in the output folder that is stored in yolov8

First collection of image dataset from the nearby station with different objects ex:Tary,Helmet, Geometry, Bucket. Using the data augmentation increasing the dataset .By using the labelling technique bounding the text box for respective images using the yolo code giving the input images to the code along with the labelling and writing the yolo code and training and testing the data using the yolo code and predicting the output.

3.6 Challenges in Object Detection:

Applications of object detection have a wide extend covering autonomous driving, detecting aerial objects, content location, reconnaissance, protect operations, mechanical autonomy, facing detection, person on foot location, visual look motor, computation of question of intrigued, brand location, and numerous more. The major challenges within the object detection incorporates; (i) The inhabitance of an protest in an picture has an inborn variety such as objects in an picture may involve lion's share of the pixels i.e., 70% to 80%, or exceptionally few pixels i.e., 10% or indeed less, (ii) Preparing of moo- determination visual substance, (iii) Taking care of shifted measured different objects in an picture, (iv) Accessibility of labelled information, and (v) Taking care of covering objects in visual substance

1.Foreground-Background class imbalance:

Awkwardness or disproportion among the occasions of diverse categories can majorly influences the show execution.

2.Detection of moderately littler objects:

All the object detection algorithms will tend to perform well on bigger objects in case the demonstrate is prepared on bigger objects. Be that as it

may, these models appear poor execution on comparatively littler measured objects.

3. **Necessity of huge datasets and computational control:**

Object detection algorithms in deep learning require bigger datasets for computation, labor seriously approaches for annotations, and capable computational assets for processing. Due to the exponential increment of produced information from different sources, it has gotten to be a dull and laborintensive assignment to explain each and each question within the visual substance.

4. **Smallersizeddatasets:**

In spite of the fact that profound learning models beat conventional machine learning approaches by a extraordinary edge, they illustrate destitute execution whereas assessing on the datasets with lessoccurrences.

5. **Inaccurate localization during predictions:**



In the figure, we have an RGB image that has been separated by its three color planes — Red, Green, and Blue. There are a number of such color spaces in which images exist — Grayscale, RGB, HSV, CMYK, etc.

You can imagine how computationally intensive things would get once the images reach dimensions, say 8K (7680×4320). The role of ConvNet is to reduce the images into a form that is easier to process, without losing features that are critical for getting a good prediction. This is important when we are to design an architecture that is not only good at learning features but also scalable to massive datasets.

	Type	Filters	Size	Output
1×	Convolutional	32	3 × 3	256 × 256
	Convolutional	64	3 × 3 / 2	128 × 128
	Convolutional	32	1 × 1	
	Convolutional	64	3 × 3	
	Residual			128 × 128
2×	Convolutional	128	3 × 3 / 2	64 × 64
	Convolutional	64	1 × 1	
	Convolutional	128	3 × 3	
	Residual			64 × 64
8×	Convolutional	256	3 × 3 / 2	32 × 32
	Convolutional	128	1 × 1	
	Convolutional	256	3 × 3	
	Residual			32 × 32
	Convolutional	512	3 × 3 / 2	16 × 16
8×	Convolutional	256	1 × 1	
	Convolutional	512	3 × 3	
	Residual			16 × 16
	Convolutional	1024	3 × 3 / 2	8 × 8
4×	Convolutional	512	1 × 1	
	Convolutional	1024	3 × 3	
	Residual			8 × 8
	Avgpool		Global	
	Connected		1000	
	Softmax			

Fig 3.6.2 Darnet design

Darknet Design:

YOLO employs the Darknet design, a custom neural network architecture particularly planned for YOLO. Darknet comprises of convolutional layers taken after by max-pooling layers for highlight extraction. Darknet design is a convolutional neural network that acts as a backbone for the YOLOv3 object detection approach. The improvements upon its predecessor Darknet design include the use of residual connections, as well as more layers.

	Type	Filters	Size	Output
1×	Convolutional	32	3 × 3	256 × 256
	Convolutional	64	3 × 3 / 2	128 × 128
	Convolutional	32	1 × 1	
	Convolutional	64	3 × 3	
	Residual			128 × 128
2×	Convolutional	128	3 × 3 / 2	64 × 64
	Convolutional	64	1 × 1	
	Convolutional	128	3 × 3	
	Residual			64 × 64
8×	Convolutional	256	3 × 3 / 2	32 × 32
	Convolutional	128	1 × 1	
	Convolutional	256	3 × 3	
	Residual			32 × 32
8×	Convolutional	512	3 × 3 / 2	16 × 16
	Convolutional	256	1 × 1	
	Convolutional	512	3 × 3	
	Residual			16 × 16
4×	Convolutional	1024	3 × 3 / 2	8 × 8
	Convolutional	512	1 × 1	
	Convolutional	1024	3 × 3	
	Residual			8 × 8
	Avgpool		Global	
	Connected		1000	
	Softmax			

Fig 3.6.2 Darknet design

Convolutional Layers:

Darknet contains numerous convolutional layers with channels of different sizes to extricate highlights from the input picture. These layers capture spatial data at distinctive scales. Convolutional Neural Network (ConvNet/CNN) is a Deep Learning algorithm that can take in an input image, assign importance (learnable weights and biases) to various aspects/objects in the image, and be able to differentiate one from the other. The pre-processing required in a ConvNet is much lower as compared to other classification algorithms. While in primitive methods filters are hand-engineered, with enough training, ConvNets have the ability to learn these filters/characteristics.

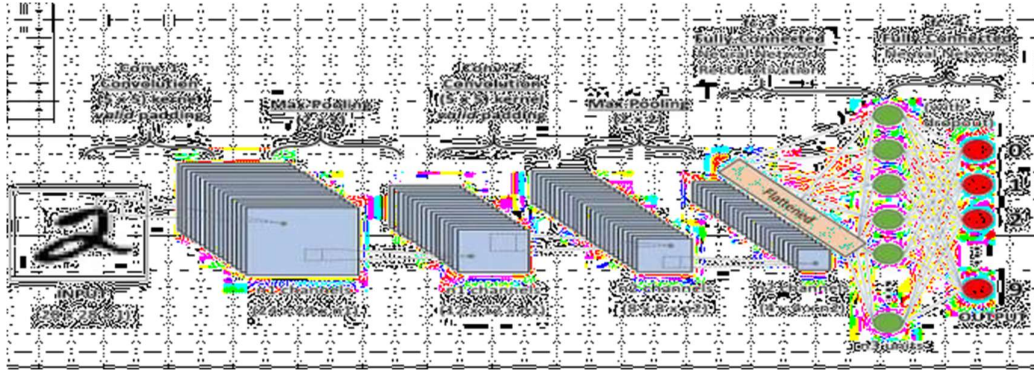
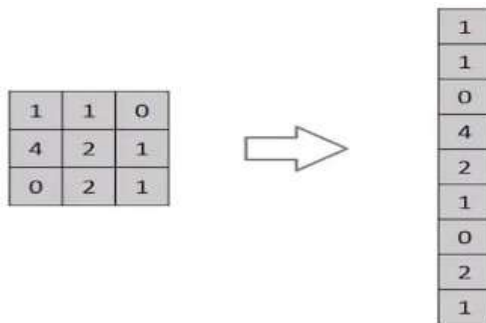


Fig 3.6.3 convolutional layers

The architecture of a ConvNet is analogous to that of the connectivity pattern of Neurons in the Human Brain and was inspired by the organization of the Visual Cortex.

Why ConvNets over Feed-Forward Neural Nets?



Flattening of a 3x3 image matrix into a 9x1 vector

An image is nothing but a matrix of pixel values, right? So why not just flatten the image (e.g. 3x3 image matrix into a 9x1 vector) and feed it to a Multi-Level Perceptron for classification purposes? Uh.. not really.

In cases of extremely basic binary images, the method might show an average precision score while performing prediction of classes but would have little to no accuracy when it comes to complex images having pixel dependencies throughout.

A ConvNet is able to successfully capture the Spatial and Temporal dependencies in an image through the application of relevant filters. The architecture performs a better fitting to the image dataset due to the reduction

in the number of parameters involved and the reusability of weights. In other words, the network can be trained to understand the sophistication of the image better.

Convolution Layer — The Kernel

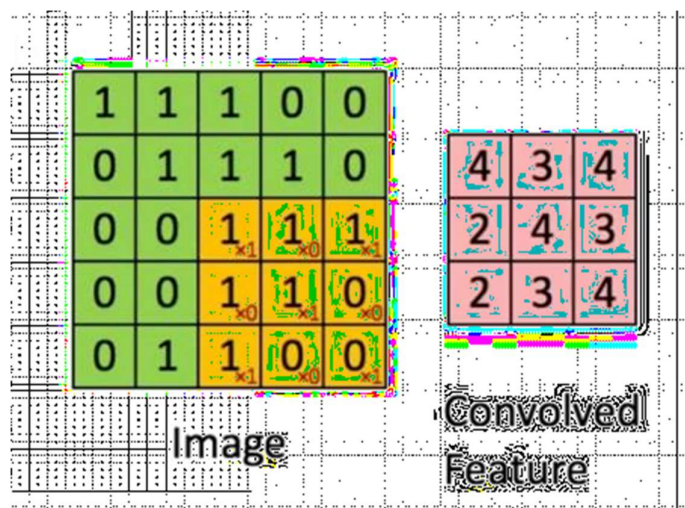


Fig 3.6.4 the kernel

Convoluting a $5 \times 5 \times 1$ image with a $3 \times 3 \times 1$ kernel to get a $3 \times 3 \times 1$ convolved feature Image Dimensions = 5 (Height) x 5 (Breadth) x 1 (Number of channels, eg. RGB)

In the above demonstration, the green section resembles our $5 \times 5 \times 1$ input image, I. The element involved in the convolution operation in the first part of a Convolutional Layer is called the Kernel/Filter, K, represented in color yellow. We have selected K as a $3 \times 3 \times 1$ matrix.

Kernel/Filter, K =

1 0 1

0 1 0

1 0 1

The Kernel shifts 9 times because of Stride Length = 1 (Non-Strided), every time performing an elementwise multiplication operation (Hadamard Product) between K and the portion P of the image over which the kernel is hovering.

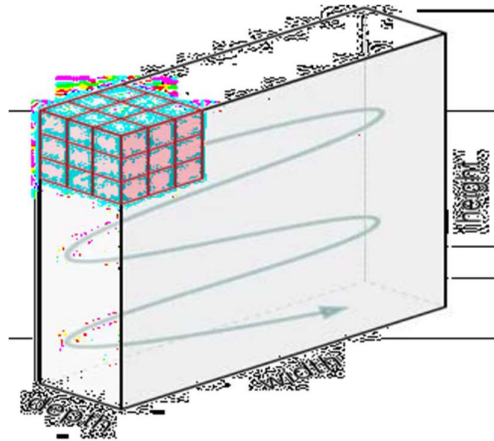


Fig 3.6.5 Movement of the Kernel

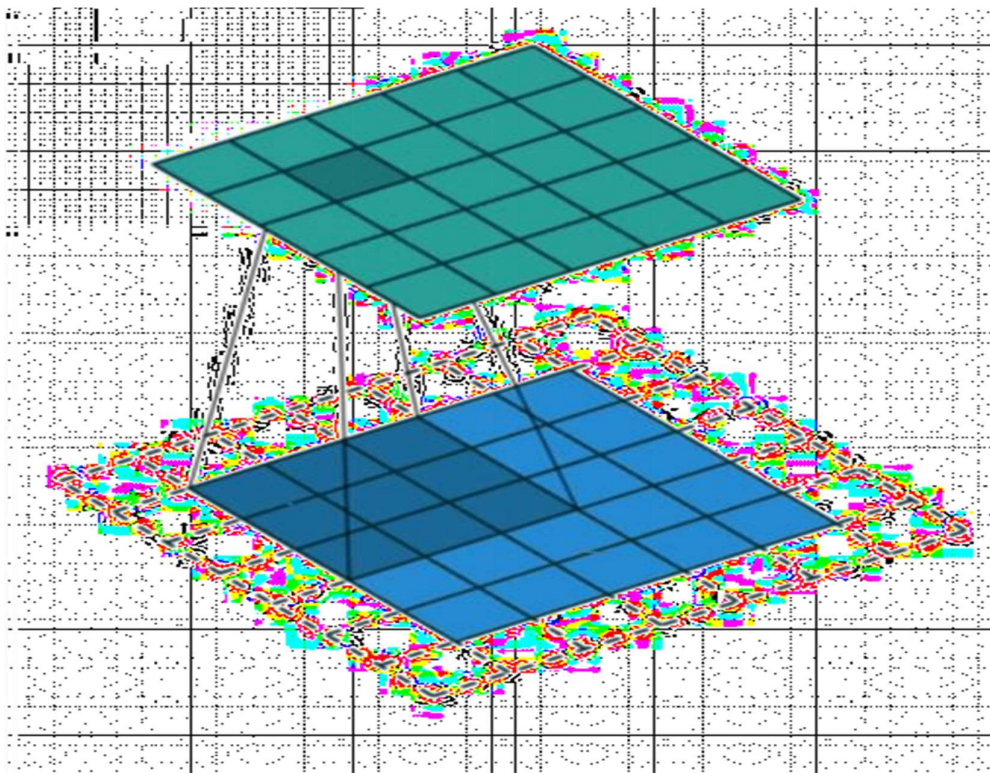
The filter moves to the right with a certain Stride Value till it parses the complete width. Moving on, it hops down to the beginning (left) of the image with the same Stride Value and repeats the process until the entire image is traverse.

When we augment the $5 \times 5 \times 1$ image into a $6 \times 6 \times 1$ image and then apply the $3 \times 3 \times 1$ kernel over it, we find that the convolved matrix turns out to be of

dimensions $5 \times 5 \times 1$. Hence the name — Same Padding

On the other hand, if we perform the same operation without padding, we are presented with a matrix that has dimensions of the Kernel ($3 \times 3 \times 1$) itself — Valid Padding. The following repository houses many such GIFs which would help you get a better understanding of how Padding and Stride Length work together to achieve results relevant to our needs.

Similar to the Convolutional Layer, the Pooling layer is responsible for reducing the spatial size of the Convolved Feature. This is to decrease the computational power required to process the data through dimensionality reduction. Furthermore, it is useful for extracting dominant features which are rotational and positional invariant, thus maintaining the process of effectively training the model.

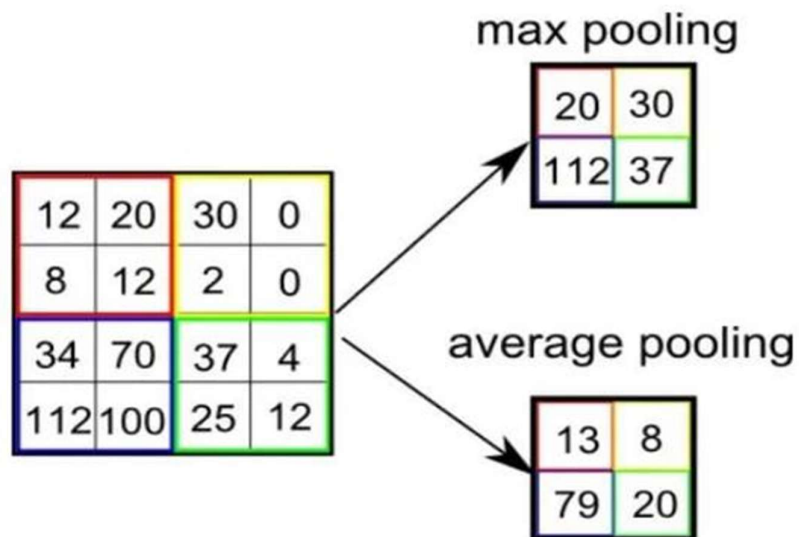


Downsampling:

Downsampling layers (regularly utilizing walk in convolutional layers or max-pooling) are utilized to decrease the spatial measurements of include maps where as expanding their profundity. There are two types of Pooling: Max Pooling and Average Pooling. Max Pooling returns the maximum value from the portion of the image covered by the Kernel. On the other hand, Average Pooling returns the average of all the values from the portion of the image covered by the Kernel.

Max Pooling also performs as a Noise Suppressant. It discards the noisy activations altogether and also performs de-noising along with dimensionality reduction. On the other hand, Average Pooling simply performs dimensionality reduction as a noise-suppressing mechanism. Hence, we can say that Max Pooling performs a lot better than Average Pooling.

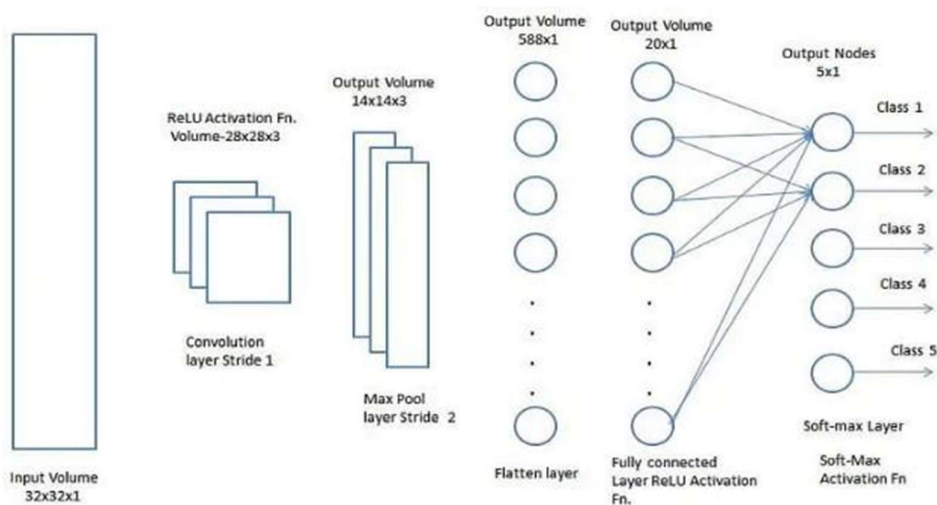
Pooling Layer



The Convolutional Layer and the Pooling Layer, together form the i -th layer of a Convolutional Neural Network. Depending on the complexities in the images, the number of such layers may be increased for capturing low-level details even further, but at the cost of more computational power.

After going through the above process, we have successfully enabled the model to understand the features. Moving on, we are going to flatten the final output and feed it to a regular Neural Network for classification purposes.

Fully Connected Layer (FC Layer)



Adding a Fully-Connected layer is a (usually) cheap way of learning non-linear combinations of the high-level features as represented by the output of the convolutional layer. The Fully-Connected layer is learning a possibly non-linear function in that space.

Now that we have converted our input image into a suitable form for our Multi-Level Perceptron, we shall flatten the image into a column vector. The flattened output is fed to a feed-forward neural network and backpropagation is applied to every iteration of training. Over a series of epochs, the model is able to distinguish between dominating and certain low-level features in images and classify them using the Softmax Classification technique.

The feature maps are smoothed and passed through completely associated layers to extricate high-level highlights and spatial connections.

Detection Head:

YOLO's detection head could be a single convolutional layer that simultaneously predicts different bounding boxes and their corresponding probabilities over the whole picture. Each bounding box forecast comprises of facilitates (center coordinates, width, tallness) certainty scores

Grid Cells:

The input image is divided into a grid of cells. Each cell is responsible for predicting bounding boxes that have their center within the cell.

Anchor Boxes:

YOLO uses anchor boxes to predict multiple bounding boxes of different sizes and aspect ratios for each grid cell. These anchor boxes are pre-defined based on the dataset.

Objectness Score:

YOLO predicts an "objectness" score for each bounding box, indicating the confidence that the box contains an object.

3.7 The Evolution of YOLO: Object detection algorithms Contents

Importance of real-time object detection

Introduction to the YOLO algorithm (You Only Look Once) and its significance in the field
Pre-YOLO era: A glimpse into traditional object detection

The birth of YOLO: Simultaneous bounding boxes classification and regression problem
YOLO v1: The first version

YOLO v2 (YOLO9000): Improvements and innovations

Non-Maximum Suppression (NMS): After predictions are made, NMS is applied to remove duplicate detections and refine the final set of detected objects.

Output:

The output of YOLO is a set of bounding boxes along with their corresponding class labels and confidence scores for the detected objects.

3.8 The Evolution of YOLO: Object detection algorithms Contents

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YOLO v3: Aiming for balance
YOLO v4: The quest for efficiency

YOLO v5: Streamlined for production
Recent advancements in the YOLO family

Other object detection algorithms inspired by YOLO
Key insights

Object detection is a pivotal aspect of computer vision that involves identifying and locating specific objects within an image or video frame. Unlike image classification tasks, assigning a singular label to an entire image, object detection algorithms discern multiple objects in an image and assign each a bounding box, indicating its position within the input image.

This task is typically powered by deep learning models such as convolutional neural networks (CNNs), which are trained on vast datasets to recognize patterns and features of different objects. Object detection algorithms have diverse applications ranging from self-driving cars, where it's crucial to detect other vehicles or pedestrians, to retail, where it can automatically track products on shelves.

Over the years, numerous algorithms like R-CNN, SSD, and YOLO have been developed to optimize accuracy and inference speed, making object detection a dynamic and rapidly evolving field in computer vision.

Importance of real-time object detection:

Real-time object detection elevates the capabilities of computer vision by enabling instantaneous recognition and localization of objects within a dynamic environment. The essence of "real-time" lies in processing images or video frames as they arrive, typically at a rate of several frames per second, without noticeable delays.

This immediacy is critical in applications where split-second decisions are paramount. For instance, in autonomous vehicles, the ability to detect and react to obstacles, traffic signals, or pedestrians in real time can be the difference between safe navigation and potential accidents.

Similarly, in security and surveillance, real-time detection can trigger immediate alerts for suspicious activities, enhancing safety measures. Moreover, augmented reality (AR) applications rely on it to seamlessly overlay digital information on the physical world. As technological advancements continue, the demand for real-time object detection is expected to grow, underlining its significance in shaping a responsive and interactive digital future.

Yolo detection:

Introduction to the YOLO algorithm (You Only Look Once) and its significance in the field

The YOLO algorithm, which stands for "You Only Look Once," revolutionized the realm of object detection with its distinctive approach. Unlike traditional methods involving multiple stages, such as region proposals followed by classification, YOLO accomplishes detection in a single forward pass of the network. By dividing an input image into a grid and predicting multiple bounding boxes and class probabilities simultaneously for each grid cell, the YOLO algorithm offers both speed and accuracy.

Yolo object detection:

This unified architecture allows it to process images remarkably fast, making it particularly suited for real-time applications. Since its inception, several versions of the YOLO model have been released, each refining its performance and architecture. Its efficiency in balancing speed and accuracy underscores YOLO's significance in the field.

Pre-YOLO era: A glimpse into traditional object detection:

Before the advent of YOLO, object detection predominantly relied on methods such as the sliding window and region proposal techniques. The sliding window approach involved scanning the entire image with varying-sized windows to detect objects at different scales and locations. This method, although comprehensive, was computationally intensive as it required evaluating numerous windows across the input image.

On the other hand, region proposal techniques, as seen in methods like R-CNN, first identified regions of interest that likely contained objects and then classified each proposed region. While this reduced the number of evaluations compared to the sliding window, it still consisted of a two-step process that could be slow and computationally demanding.

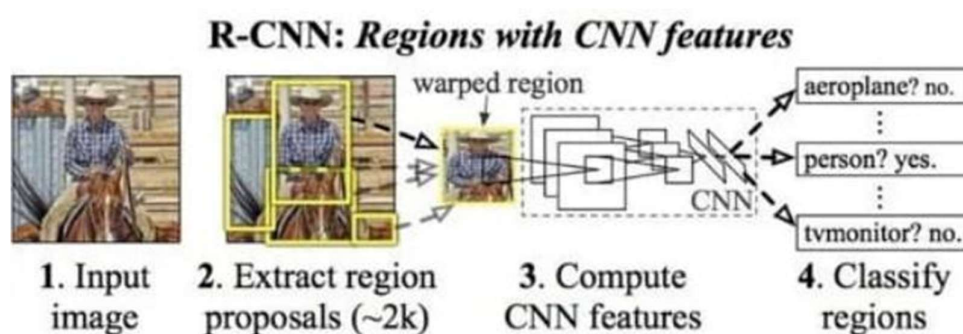


Fig 3.7.1 cnn features

Both these strategies faced challenges in terms of speed, especially when applied to high-resolution images or real-time video streams. Furthermore, striking a balance between accuracy and computational efficiency was a constant struggle. The need for faster and more efficient algorithms that didn't compromise on accuracy paved the way for breakthroughs like the

YOLO model, which addressed many of these pre-existing challenges.

The birth of YOLO: Simultaneous bounding boxes classification and regression problem

The emergence of YOLO marked a transformative moment in the landscape of object detection. Breaking away from traditional paradigms, YOLO introduced a novel methodology underpinned by a few core principles.

Firstly, YOLO's expansion, "You Only Look Once," encapsulates its foundational idea: object detection should occur in a single forward pass through the neural network.

This sharply contrasts with methods that separate object localization and classification into distinct stages.

YOLO's strategy divides an image into a grid, and for each grid cell, it predicts multiple bounding boxes and their associated class probabilities simultaneously. This unified framework drastically reduces computational overhead, making real-time detection feasible without sacrificing accuracy.

Compared to previous methods, YOLO's streamlined approach offers significant advantages.

It reduces latency, making it apt for real-time applications, and its holistic view of the image ensures contextual understanding, reducing false positives. With YOLO's inception, the field witnessed a paradigm shift, emphasizing both efficiency and accuracy in object detection.

YOLO v1: The first version

YOLO v1 burst onto the computer vision scene and introduced a fresh perspective on object detection. Its architecture was distinct; the model split images into a 7×7 grid, and for each grid cell, predicted bounding boxes along with class probabilities, ultimately producing a fixed number of predictions. This was inherently different from the iterative proposals and classifications of prior methods.

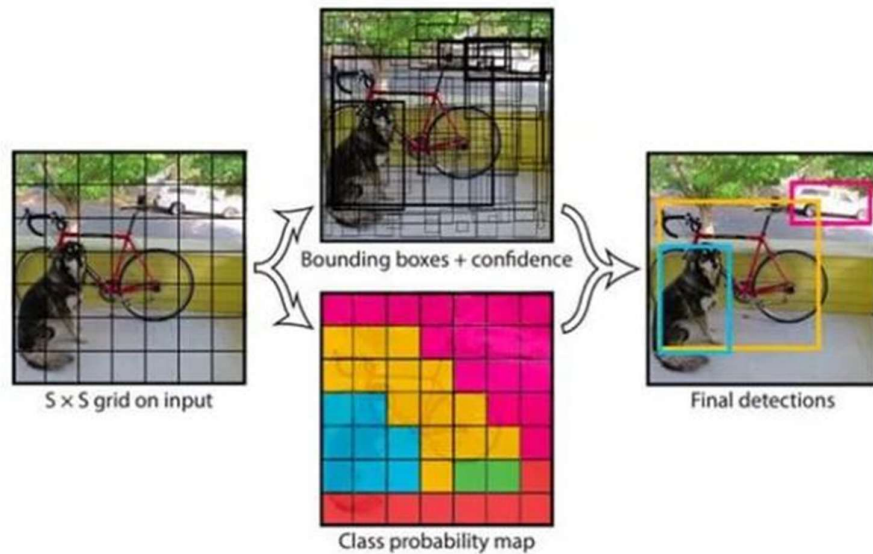


Fig 3.7.2 yolov1

Performance-wise, YOLO v1 outstripped many contemporaneous algorithms in terms of speed, making real-time detection not just a theoretical aspiration but a tangible reality. However, its first iteration had limitations. Detecting smaller objects proved challenging for YOLO v1, as its design was more inclined towards identifying larger, more prominent objects.

Additionally, the model demonstrated sensitivity to the positioning of objects. If an object straddled multiple grid cells, the model occasionally faltered in its predictions. Despite these initial shortcomings, YOLO v1 laid a robust foundation, setting the stage for subsequent refinements and iterations that would further refine the YOLO model and elevate its

capabilities.

YOLO v2 (YOLO9000): Improvements and innovations

YOLO v2 model suggested several improvements on top of the v1 architecture, such as multi-scale training, anchor boxes, and the Darknet-19 architecture.

Multi-scale training

YOLO v2, often referred to as "YOLO9000" due to its ability to detect over 9000 object classes, introduced several refinements to address the shortcomings of its predecessor. One of the most significant enhancements was the concept of multi-scale training. Recognizing that objects in real-world images appear at various scales and sizes, the YOLO v2 model was trained to predict objects across different scales.

In practical terms, during the training process, the size of the input images was periodically changed. For instance, the network might be trained on images of size 320x320 for a certain number of batches, and then the resolution would be increased to, say, 416x416 for the subsequent ones. This process continued for even larger resolutions. This dynamic adjustment enabled the model to better adapt to objects of varying dimensions.

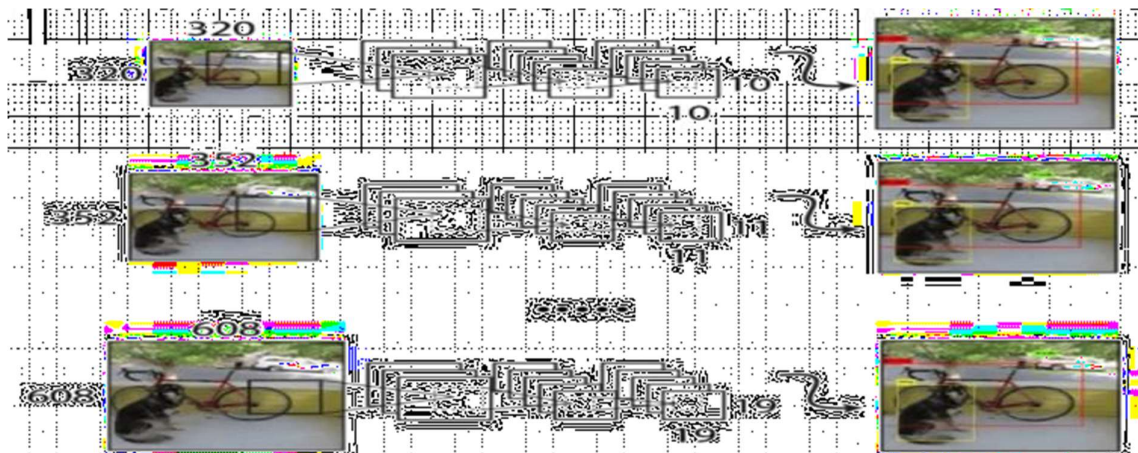


Fig 3.7.3 multi scale training

This approach has two benefits. Firstly, it significantly improved the model's ability to detect smaller objects, a challenge that YOLO v1 faced. Secondly, it endowed the model with a more generalized understanding of object scales, making it more robust and versatile across diverse real-world scenarios. This multi-scale training strategy was a key factor in the improved performance and versatility of YOLO v2 over the original version.

Anchor boxes

Another groundbreaking enhancement in YOLO v2 was the introduction of anchor boxes. Recognizing that objects in images come in various shapes and aspect ratios, the anchor box concept aimed to improve the bounding box coordinates prediction mechanism.

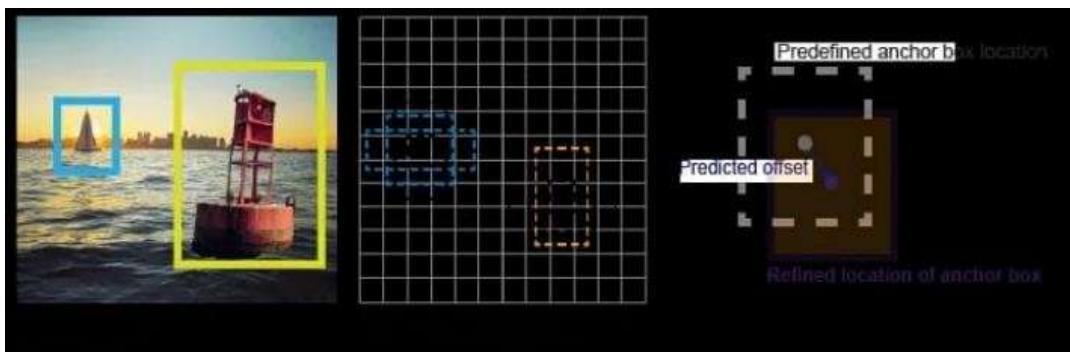


Fig 3.7.4 anchor boxes

Predicting small bounding boxes from scratch can be challenging in traditional object detection approaches. Anchor boxes, however, provide predefined bounding box coordinates, fixed aspect ratios, and sizes based on common shapes found in the training dataset. During the prediction phase, the model adjusts these anchor boxes to better fit the actual bounding boxes of the objects in the image.

The idea is that it's easier for the model to predict adjustments to these predefined bounding boxes than to generate accurate bounding boxes from scratch. For example, if a dataset frequently contains tall, rectangular objects (like standing humans), one of the anchor boxes might have a tall, thin shape. During detection, the model can use this anchor box as a starting point and make adjustments to fit each detected person better.

Incorporating anchor boxes was pivotal in addressing overlapping objects and improving recall, especially for objects that might have been challenging due to their aspect ratios or sizes. By utilizing anchor boxes, YOLO v2 demonstrated a more refined bounding box prediction, enhancing its detection accuracy and making it even more competitive in the object detection landscape.

Darknet-19

Darknet-19 is the underlying neural network architecture that YOLO v2, or YOLO9000, utilizes for its object detection tasks. It's a custom-built architecture designed to be fast and efficient, which is why it's particularly well-suited for real-time applications.

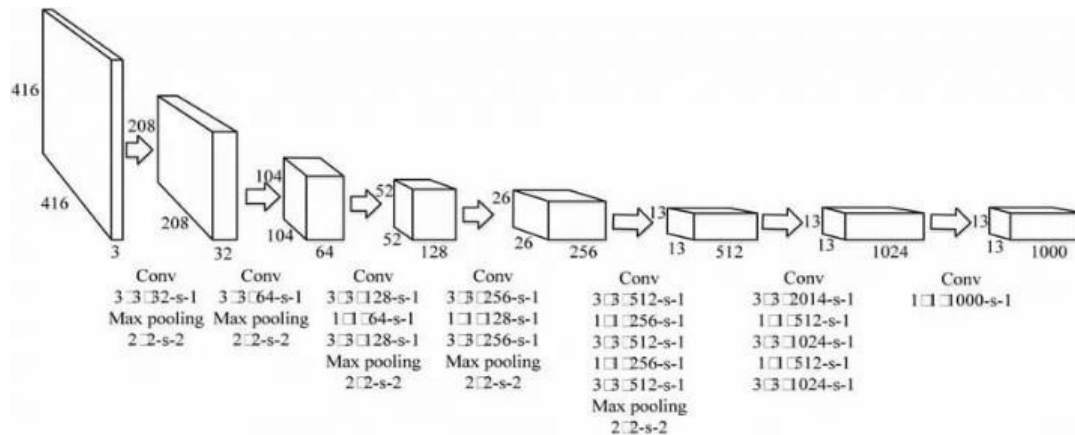


Fig 3.7.5 darnet 19

Here are some key features of the Darknet-19 architecture:

Layers

Darknet-19 comprises 19 convolutional layers accompanied by five max-pooling layers. The name "Darkneet-19" originates from these 19 convolutional layers.

Convolutional Layers

These layers make use of 3x3 filters primarily, which helps in retaining fine-grained features. Some of the later layers in the architecture use 1x1 filters for dimension reduction.

Activation Functions:

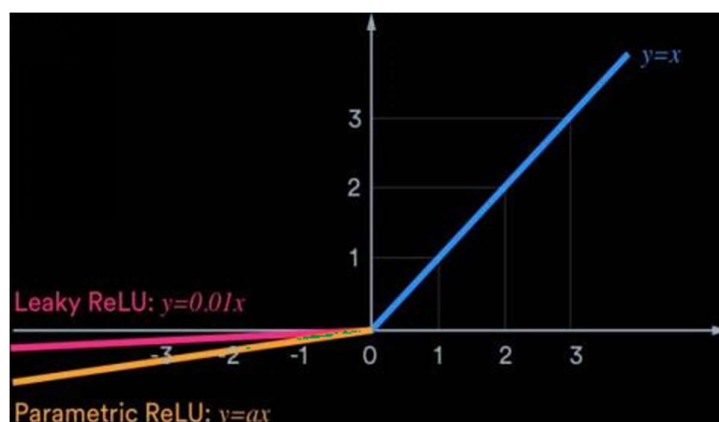


Fig 3.7.6 activation functions

Each convolutional layer is followed by a non-linear activation function - Leaky ReLU (Rectified Linear Unit), which helps add non-linearity to the model without significant computational overhead.

Normalization:

Darknet-19 incorporates batch normalization after its convolutional layers. This aids in stabilizing training, reducing the need for meticulous initialization, and somewhat mitigating the effects of overfitting.

Global average pooling:

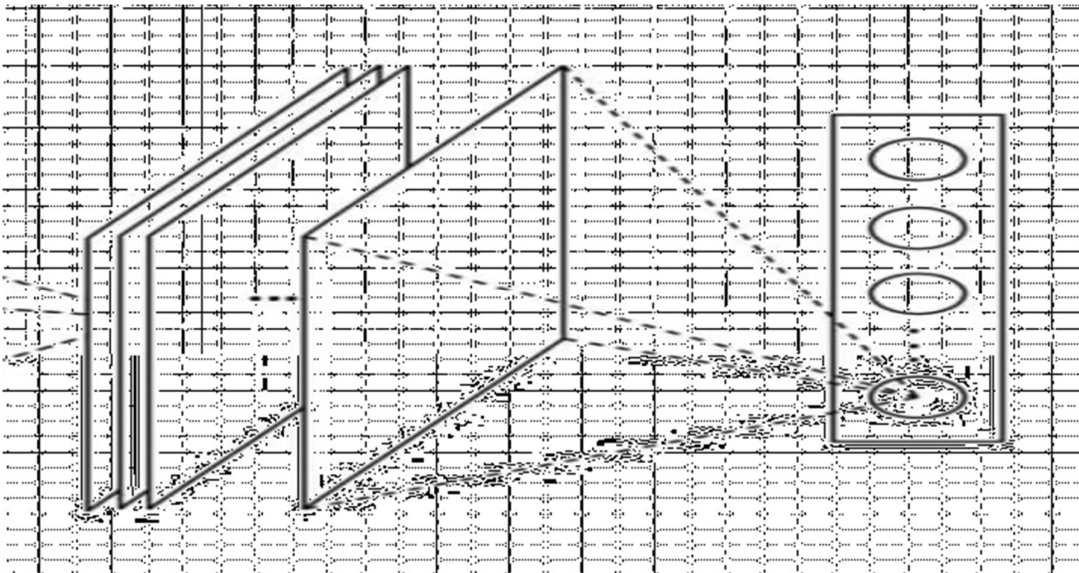


Fig 3.7.7 global average pooling

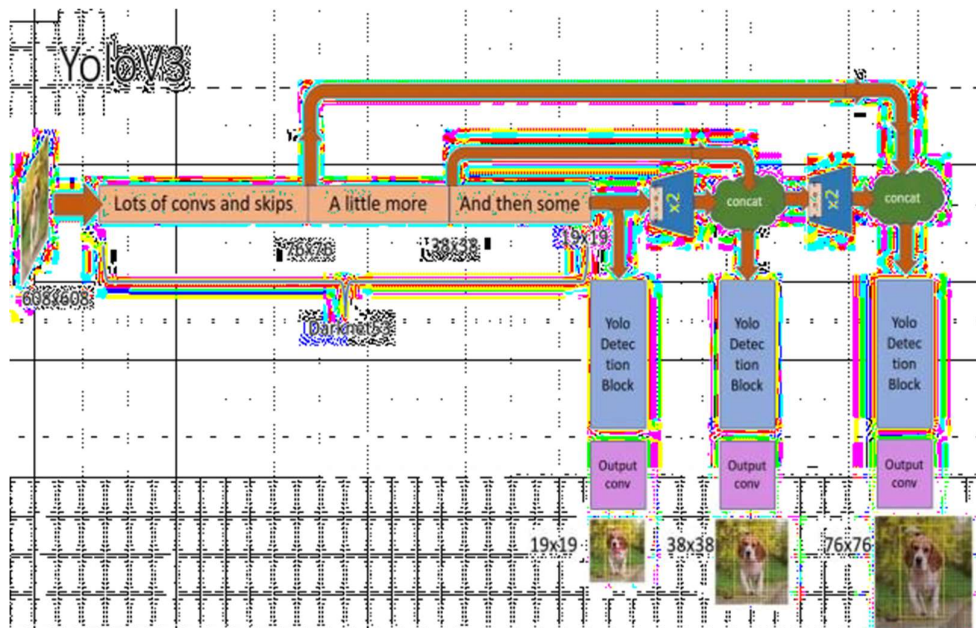
At the end of the network, before the final softmax layer, global average pooling is used. This replaces the fully connected layers commonly seen in other architectures, leading to a reduction in the overall number of parameters and, consequently, less risk of overfitting.

The choice of Darknet-19 for YOLO v2 wasn't arbitrary. Its design is a testament to the requirement of having a high-performing yet computationally efficient neural network for real-time object detection, perfectly aligning with the objectives of the YOLO series.

YOLO v3: Aiming for balance:

As object detection continued to evolve, the need for a model that strikes a harmonious balance between detection speed and accuracy became paramount. YOLO v3, the third iteration of the renowned YOLO series, emerged as an answer to this call. While retaining the foundational principles of its predecessors, YOLO v3 introduces enhancements and refinements, fine-tuning its performance across varying object sizes and ensuring a more holistic detection capability. In this section, we delve into the intricacies of YOLO v3,

exploring its architectural advancements and understanding its drive toward achieving a balanced object detection model.



Modifications in architecture

YOLO v3 brought about several architectural modifications to further refine object detection capabilities. Here's a breakdown of these changes:

Darker architecture

YOLO v3 uses a deeper architecture named "Darknet-53", which combines the best of Darknet-19 (used in YOLO v2) and some of the residual connections from the ResNet architecture. This resulted in 53 convolutional layers, hence the name. These layers provide a richer feature set for detection while the residual connections assist in training the deeper network.

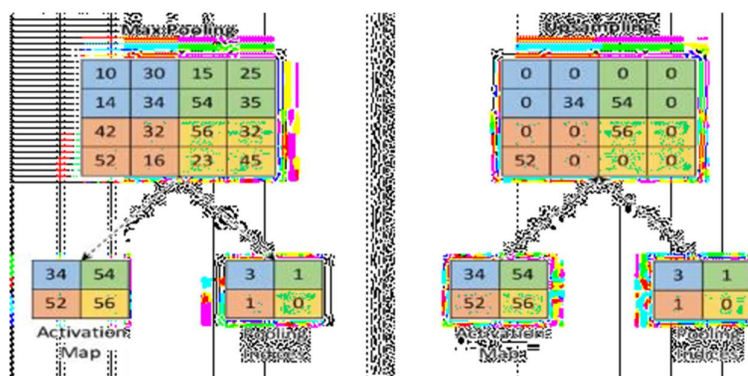
Three scales for detection Instead of a single scale, YOLO v3 makes detections at three different scales. This is accomplished using three different sizes of anchor boxes for each scale, thus effectively addressing the issue of detecting objects of varying sizes.

Use of three YOLO layers:

To be aligned with the three scales, YOLO v3 introduces three YOLO layers in its network. Each layer is responsible for detecting objects at a particular scale, ensuring a more robust multi-scale object detection.

Bounding box prediction

Instead of using the width and height directly for predicting bounding boxes, YOLO v3 uses a logarithmic space to predict bounding box sizes. This aids in improving accuracy, especially for larger objects.



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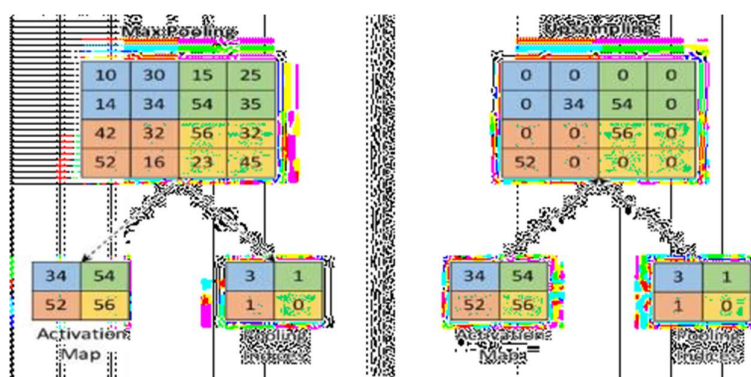
Class prediction:

Instead of using a softmax layer for class prediction (as in YOLO v2), YOLO v3 employs independent logistic classifiers. This allows the model to predict multiple object classes for the same bounding box, making it capable of handling overlapping objects of different classes.

Improved loss function:

The loss function was modified to better balance the trade-off between localization errors of bounding box coordinates and classification errors (correct object category identification).

Use of upsampling



YOLO v3 uses upsampling layers to merge features from previous layers, providing a richer feature set for detection at different scales. This is especially beneficial for detecting smaller objects.

In essence, the architectural modifications in YOLO v3 were focused on enhancing the model's precision without compromising its speed. The shift towards multi-scale detection, combined with the deeper and richer Darknet-53 backbone, allowed YOLO v3 to achieve a commendable balance between performance and accuracy in object detection tasks.

ation without significantly increasing computational demands. YOLO v3 And SOTA models:

During its release, YOLO v3 was juxtaposed against several state-of-the-art models to benchmark its performance. Here's a comparative analysis:

Speed vs. accuracy trade-off

One of the most distinct attributes of YOLO v3 was its ability to maintain a compelling balance between speed and accuracy. While models like Faster R-CNN and RetinaNet might have achieved slightly superior accuracy on certain benchmarks, they typically did so at the cost of computational speed. YOLO v3, on the other hand, retained commendable accuracy while operating at real-time or near-real-time speeds, a balance many models struggled with.

Detection across scales:

With its multi-scale detection, YOLO v3 showcased improved performance

in detecting smaller objects compared to its predecessor, YOLO v2. However, models like SSD (single shot multibox detector) and some configurations of RetinaNet, which also incorporated multi-scale detection, were competitive in this regard.

Generalization:

YOLO v3's architecture, especially with the Darknet-53 backbone and the use of logistic classifiers for class prediction, provided it with a better generalization capability. This meant that YOLO v3 was more adept at handling varied real-world scenarios than some models, which might have been overly optimized for specific benchmark test data.

Limitations:

Like any model, YOLO v3 had its challenges. For instance, densely packed objects or objects with overlapping bounding boxes sometimes pose detection challenges. Some other models, especially those optimized for higher average precision (AP) at the cost of speed, might outperform YOLO v3 in such intricate scenarios.

Versatility:

The design philosophy behind YOLO always emphasized broad applicability – a model that can detect thousands of object types in varied environments. In benchmarks that evaluated this versatility, such as detecting uncommon object types, YOLO v3 often outshone models that were narrowly optimized for specific, common object categories.

In summary, while other state-of-the-art models of the time might have excelled in specific niches or under certain conditions, YOLO v3 was celebrated for its all-roundedness. It provided a unique combination of speed, accuracy, and versatility, making it a preferred choice for various real-world applications, from surveillance systems to autonomous vehicles.

YOLO v4:

The quest for efficiency, As the relentless march of innovation in object detection progressed, the YOLO series presented its fourth iteration, YOLO v4, with a clear objective: achieving unparalleled efficiency. While the

preceding versions had set benchmarks in balancing speed and accuracy, YOLO v4 sought to redefine these boundaries, optimizing both computational efficiency and performance of bounding box detection. This version embodies the culmination of learnings from past iterations and integrates novel techniques from contemporary research. In this section, we'll unravel the intricate advancements of YOLO v4 and explore how it raised the bar in the domain of efficient object detection.

Focus on efficiency and scalability

YOLO v4's design ethos is deeply rooted in efficiency and scalability, ensuring that the model not only achieves state-of-the-art detection performance but is also optimized for deployment in various scenarios, ranging from high-end servers to edge devices.

Optimized backbones:

YOLO v4 leverages the CSPDarknet53 architecture, which, as previously discussed, provides an efficient gradient flow. This design choice optimizes the model's training process, ensuring a more rapid convergence while demanding fewer computational resources than other sophisticated backbones.

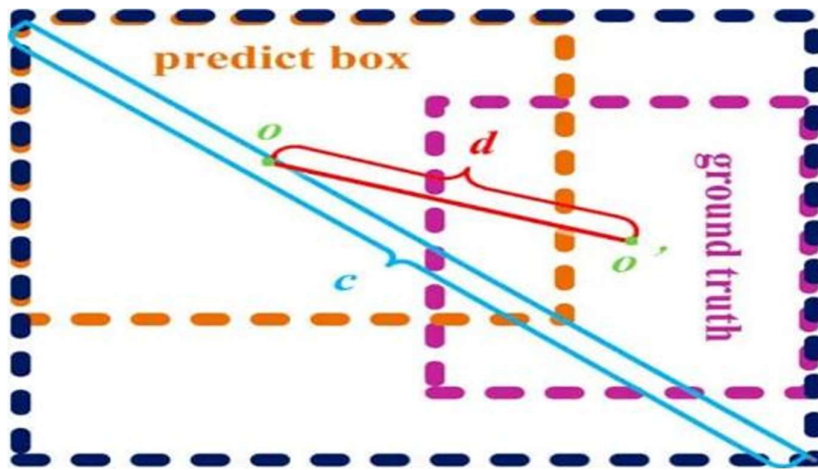
Mish activation function



Fig 3.7.8 mish activation function

YOLO v4 introduces the mish activation function, which has been shown to yield better performance than traditional activation functions like ReLU while maintaining a comparable computational cost.

CIOU Loss:



The use of the complete intersection over union (CIOU) loss aids in better bounding box regression, leading to more precise object localization.

Modularity and scalability

One of the core strengths of YOLO v4 lies in its modular design. Various components, like PANet or SAM block, can be added or removed based on the computational budget and the application's specific requirements. This makes YOLO v4 highly scalable – a critical feature when deploying the model on devices with varying computational capacities.

Advanced data augmentation:

YOLO v4 integrates advanced data augmentation techniques like Mosaic augmentation and CutMix. These techniques not only improve detection performance but also enhance the model's robustness. A more robust model can be deployed in diverse real-world scenarios without the need for excessive fine-tuning, which is a boon for efficiency.

The "neck" of the architecture, which bridges the backbone and the final detection layers (or "head"), is designed to merge features across different scales efficiently. The detection head is fine-tuned for speed, with

optimizations ensuring rapid and accurate bounding box and class predictions.



a) Sample images for Mosaic augmentation



b) Sample images for CutMix augmentation

Attention mechanisms

By focusing on relevant parts of the image through mechanisms like the SAM block, YOLO v4 ensures that computational resources are spent on regions of importance. This selective attention translates to more efficient processing, especially in cluttered or diverse scenes.

Innovations in YOLO v4:

CSPDarknet53, PANet, and SAM Block

YOLO v4 introduced a series of architectural and methodological advancements to refine its object detection capabilities further. Among these, CSPDarknet53, PANet, and the SAM block stand out as key components that significantly bolstered its performance.

Neck and head optimizations:

The "neck" of the architecture, which bridges the backbone and the final detection layers (or "head"), is designed to merge features across different scales efficiently. The detection head is fine-tuned for speed, with optimizations ensuring rapid and accurate bounding box and class predictions.

In essence, every layer and component in YOLO v4 is meticulously designed or chosen to push the boundaries of efficiency. Whether it's the choice of backbone, the activation functions, or the attention mechanisms, the model consistently strives for a balance between performance and computational cost, making it one of the most efficient and scalable object detectors of its time.

Innovations in YOLO v4:

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CSPDarknet53:

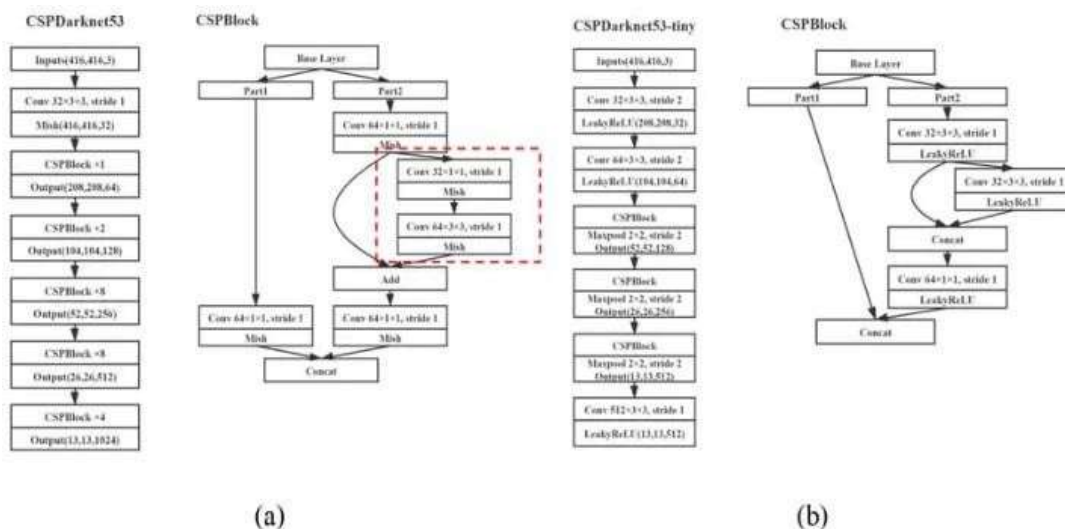


Fig 3.7.9 CSP darknet53

The "CSP" in CSPDarknet53 stands for "Cross Stage Hierarchical Networks." This technique divides the feature map into two parts and then merges them during the latter stages. Such partitioning and cross-stage hierarchy improve gradient flow, which can enhance learning, especially in deeper networks.

By integrating the CSPNet strategy with the Darknet53 architecture (which was pivotal in YOLO v3), CSPDarknet53 was born. This new backbone provides enhanced performance with fewer computations, achieving the goal of improved efficiency.

PANet (Path Aggregation Network):

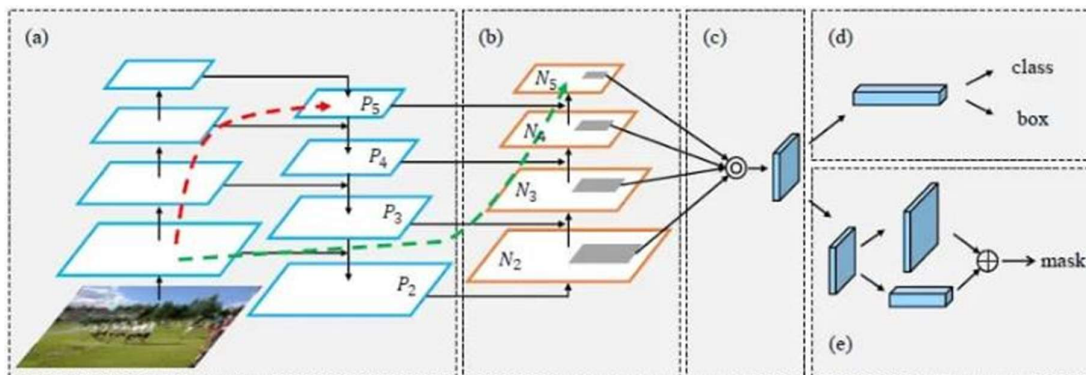
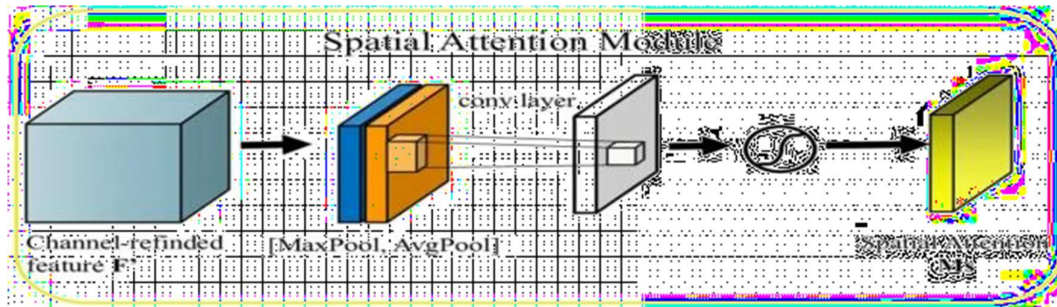


Fig 3.7.10 PA Net

PANet enhances the feature hierarchy of the network. It allows for the upward path (where semantically stronger, lower-resolution features are present) to propagate to the downward path (where higher-resolution, less semantically strong features exist).

By leveraging PANet, YOLO v4 achieves better object localization, especially for smaller objects. This architecture ensures that rich feature information is available across scales, providing a comprehensive multi-scale detection capability.

SAM Block (Spatial Attention Module)



The SAM block helps the network focus on the relevant spatial regions of the feature map. It introduces spatial attention to the features, emphasizing areas that are more likely to contain meaningful information about the object. By integrating spatial attention, the network can make more accurate decisions about object locations and sizes. This not only improves detection accuracy but also aids in the precise delineation of object boundaries.

Collectively, these three key components – CSPDarknet53 for an efficient and robust feature extraction backbone, PANet for improved hierarchical feature utilization across scales, and the SAM block for sharp spatial attention – endow YOLO v4 with a formidable capability to detect objects with high precision, all while maintaining an edge in computational efficiency. This synergy of

methodologies pushes YOLO v4 closer to the ideal intersection of speed and accuracy in the object detection landscape.

CIOU loss for better bounding box prediction

Accurate bounding box prediction is paramount in object detection, as it determines how precisely an object's location and shape are delineated. Traditionally, models employed metrics like intersection over union (IoU) to measure the overlap between the predicted bounding box and the ground truth box. However, relying solely on IoU can lead to certain challenges, particularly in handling overlapping bounding boxes, different bounding box sizes, or bounding boxes that don't overlap at all. To address these nuances, YOLO v4 incorporates the complete intersection over union (CIOU) loss, significantly improving bounding box prediction.

Basics of CIOU:

The CIOU loss is an advanced version of the standard IoU loss, which considers not just the overlap between bounding boxes but also the distance between the central points of the two bounding boxes, predicted and ground truth, as well as the aspect ratio.

Central point distance

One of the components of the CIOU loss calculates the distance between the centers of the predicted bounding box and the ground truth. By incorporating this distance into the loss, the model is penalized for boxes that might have a high IoU but aren't perfectly aligned regarding object center points. This ensures better alignment of detected objects.

Aspect ratio consistency:

Another component of the CIOU loss takes into account the aspect ratio, ensuring that the predicted bounding boxes not only overlap well with the ground truth bounding boxes but also maintain a consistent shape. This is particularly important for objects with distinct elongated or irregular shapes.

Advantages over standard IoU:

By accounting for these additional geometric factors, CIOU provides a more holistic measure of bounding box prediction quality. In scenarios where two bounding boxes might have the same IoU value, CIOU can differentiate between them based on their center alignment and aspect ratio consistency, leading to more accurate and consistent predictions.

Impact on training:

Integrating CIOU loss into the training process of YOLO v4 has been shown to speed up the convergence, as the model receives more comprehensive feedback on its predictions. Faster convergence can reduce training times and resources, further emphasizing the model's focus on efficiency.

YOLO v5: Streamlined for production

With the continuous evolution of the YOLO series, YOLO v5 emerges as a testament to the pursuit of perfection in the object detection model, designed with a keen focus on production-ready deployments. This iteration prioritizes not only top-notch detection capabilities but also the practicalities of integration into real-world applications. By emphasizing factors like model size, inference speed, and adaptability, YOLO v5 bridges the gap between high-performance object detection and seamless implementation in diverse environments. In this section, we will delve into the intricacies of YOLO v5 and explore how it stands out as a model truly streamlined for production scenarios.

Architecture simplifications:

In its quest to become more production-friendly, YOLO v5 introduced various architectural simplifications that made the model leaner without compromising its detection prowess. These modifications ensured faster inference times, ease of deployment, and efficient adaptability across diverse platforms. Let's discuss some of these key architectural changes:

Reduced complexity: One of the most noticeable shifts in YOLO v5 was the reduction in the model's overall complexity. By removing or refining certain layers and components that contributed less to performance, YOLO v5

achieved a balance between model size and detection accuracy.

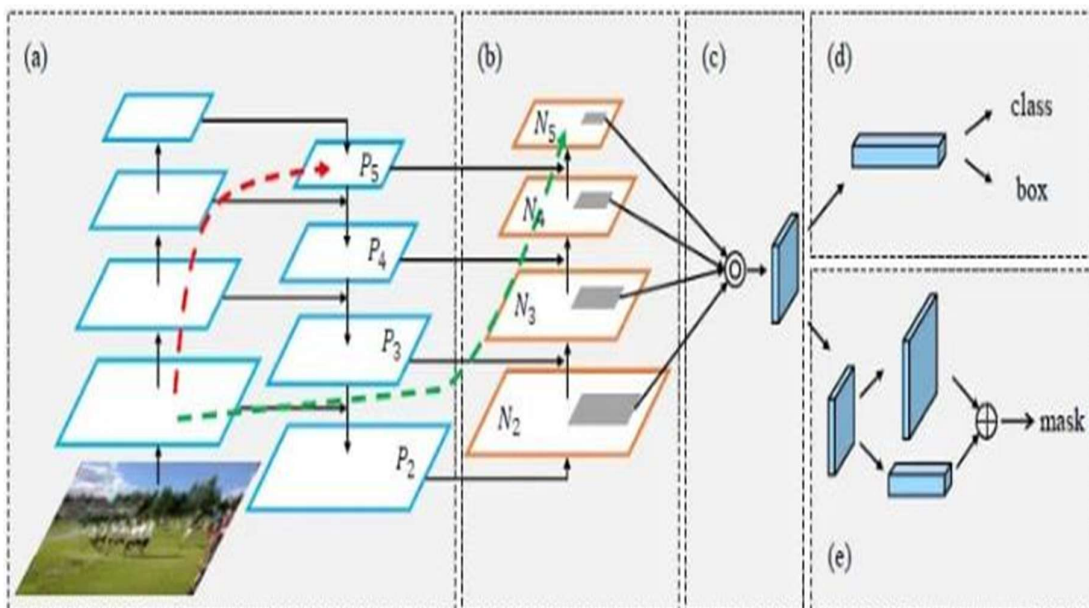
Optimized backbone: The backbone, responsible for feature extraction, underwent optimizations to ensure faster processing and reduced computational requirements. This led to quicker initial feature maps that maintained a rich input data representation.

Streamlined neck and head: The components connecting the backbone to the final output layers (often referred to as the 'neck') and the detection layers (the 'head') were streamlined. This resulted in a more efficient flow of information and faster object detection and classification processes.

Focus on edge deployment: With a growing emphasis on edge computing, YOLO v5's architecture was tailored to cater to devices with constrained computational resources. This meant optimizations that allowed for real-time inference, even on devices like smartphones or low-powered cameras.

- ❖ **Modularity:** After recognizing the diverse needs of different applications, YOLO v5's architecture was designed to be modular. This enables developers to easily add or remove components based on specific requirements, ensuring the model remains versatile across different deployment scenarios.

Efficient activations and operations: YOLO v5 incorporated activation functions and operations, which are computationally efficient and maintain



- ❖ high performance. This fine-tuning at the operation level ensured that each model part contributed effectively to the overall speed and accuracy.
- ❖ **Reduced redundancy:** Any redundancies in the architecture, such as repetitive layers or operations that contributed minimally to the overall performance, were identified and eliminated or refined. This further contributed to the model's lean nature.

In essence, YOLO v5's architectural simplifications can be seen as a meticulous exercise in efficiency. Each change, whether major or minor, was driven by the objective of creating a powerful model in its detection capabilities and primed for real-world, production-level deployments across a spectrum of devices and applications.

Emphasis on deployment and ease of use

YOLO v5, recognizing the critical need to transition from research environments to real-world applications, placed a significant emphasis on deployment readiness and user-friendliness. This shift was crucial to ensure that YOLO's powerful object detection capabilities were accessible and straightforward to implement across various platforms and scenarios. Here's a deeper look into YOLO v5's approach to deployment and ease of use:

- ❖ Pre-trained models: YOLO v5 offered a suite of pre-trained models tailored to various computational budgets. From lightweight models suitable for mobile deployments to more robust

- ❖ Tool integration: YOLO v5 emphasized compatibility with popular machine learning deployment tools, ensuring that integration into existing pipelines, whether on cloud platforms or edge devices, was smooth and hassle-free.
- ❖ Customization capabilities: After recognizing the diversity of deployment scenarios, YOLO v5 was designed to be easily customizable. This allowed developers to fine-tune the model for specific applications, ensuring optimal performance in niche environments.
- ❖ Streamlined training: For those looking to train the model on custom datasets, YOLO v5 offered an intuitive and efficient training process. This made it easier for organizations to adapt the model to their specific data and requirements.
- ❖ Robustness to varied inputs: Designed with real-world variability in mind, YOLO v5 showcased resilience to a range of input data, from low-light images to varying resolutions. This robustness made it a reliable choice for diverse deployment scenarios.
- ❖ Feedback loop integration: Acknowledging the dynamic nature of real-world applications, YOLO v5 incorporated mechanisms to easily integrate feedback loops. This allowed continuous improvement of the model based on real-world detections and adaptations.

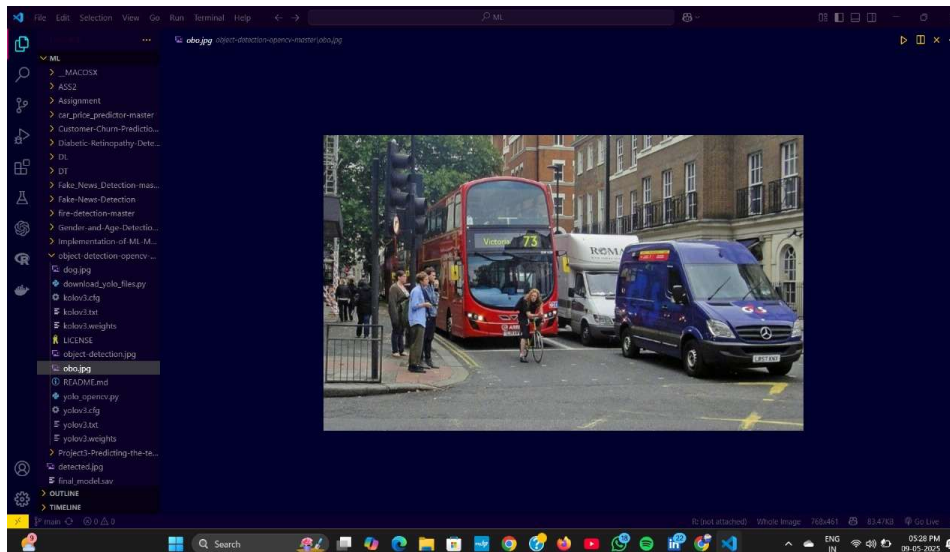
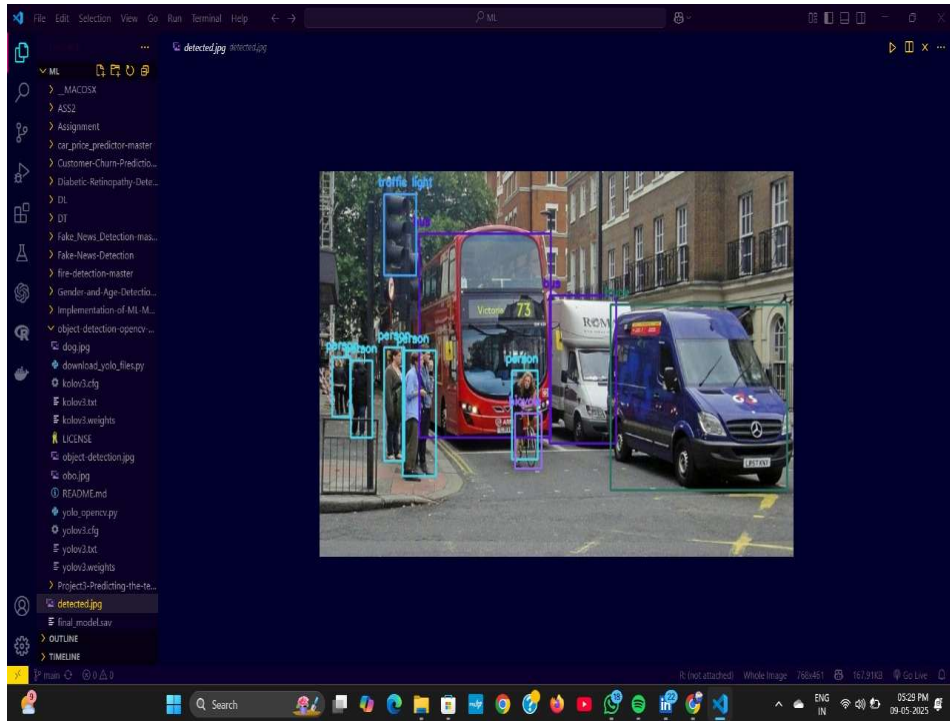
YOLO v5's commitment to deployment and ease of use highlighted a pragmatic approach, bridging the gap between cutting-edge research and tangible real-world applications. By ensuring that the model was not only powerful but also accessible and easy to integrate, YOLO v5 paved the way for widespread adoption across a myriad of industries and use cases.

Performance in real-world applications

While benchmarks and controlled evaluations provide valuable insights into a model's capabilities, the actual test for any object detection system like YOLO v5 lies in its performance in real-world scenarios. With a plethora of challenges, from varied lighting conditions to unpredictable object orientations, the real world is the ultimate proving ground. YOLO v5, with its tailored design, showcased notable prowess across diverse applications.

Let's delve into its real-world performance highlights:

Retail and inventory management: In retail environments, where object variability is high and timely inventory management is critical, YOLO v5 demonstrated precise detection capabilities. It could accurately identify and count products on shelves, aiding in stock monitoring and automated restocking processes.





Surveillance and security: In surveillance scenarios, quickly detecting anomalies or intrusions is paramount. YOLO v5's real-time processing capability and high accuracy made it a top choice for security applications, ensuring prompt alerts and responses.



Agriculture and farming: In agricultural settings, detecting pests and diseases or estimating crop yields is essential. YOLO v5's adaptability ensured accurate detections in such environments, providing farmers with valuable insights to optimize their yield and reduce losses



Traffic and urban planning:

In urban landscapes, YOLO v5 was employed to monitor traffic patterns, detect traffic violations, and analyze pedestrian movements. Its robustness to varying lighting and weather conditions ensured reliable performance around the clock.



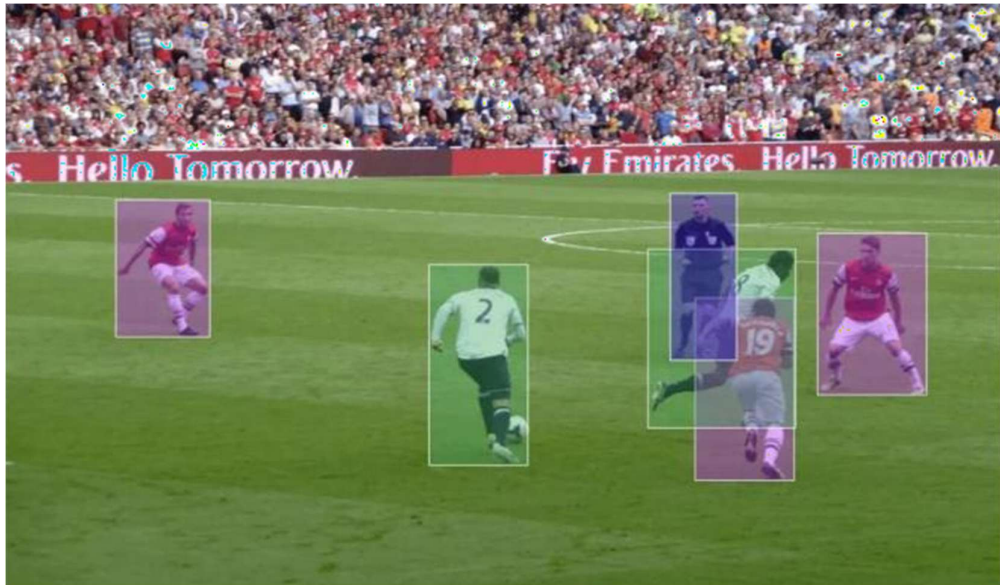
Industrial automation:

In factories and manufacturing units, YOLO v5 facilitated automated quality checks, detecting defects or misalignments on assembly lines in real time, ensuring high production standards and reduced manual interventions.



Sports analytics:

In sports broadcasts, YOLO v5 was employed to track player movements and ball trajectories, and analyze game strategies, enhancing the viewing experience and providing teams with tactical insights.



YOLO v5's performance in real-world applications underscored its holistic design philosophy. It wasn't just about achieving high scores in controlled tests but about delivering consistent, reliable, and valuable results in varied and often challenging real-world environments, affirming its stature as a leading object detection solution.

Recent advancements in the YOLO family

The YOLO family of object detection models has seen significant advancements in recent years, with each new version introducing improvements in speed, accuracy, and robustness. YOLO v6 was released in October 2021 and includes several new features and fixes, including: Two new nano models, YOLOv5n and YOLOv5n6 that have ~75% fewer parameters than previous models and can run on mobile and CPUs.

A new P5 and P6 backbone network that improves accuracy on small objects.

Several bug fixes and performance optimizations

Augmented reality (AR) and gaming: In AR applications and gaming environments, where real-time object detection melds virtual and real worlds, YOLO v5's speed and accuracy enriched user experiences, enabling more immersive interactions



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A new P5 and P6 backbone network that improves accuracy on small objects.
Several bug fixes and performance optimizations

YOLOv7 incorporates several new techniques, including:

A new backbone network called PANet that improves feature aggregation and representation
A new loss function called CIOU loss, which is more robust to object scale and aspect ratio

variations
A new anchor box mechanism called EfficientDet that reduces the number of anchor boxes needed without sacrificing accuracy

SSD is another popular single-pass object detection system, which, similar to YOLO, aims at predicting category scores and box offsets for a fixed set of default bounding boxes. While SSD's design differs in many ways, it shares YOLO's philosophy of performing detection in one forward pass of the network.

CornerNet & CenterNet: Instead of predicting bounding boxes, these models detect objects by finding their corner or center points. While their approaches are distinct, the drive to simplify and streamline detection is reminiscent of YOLO's breakthroughs.

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3.9 YOLO v8:

1. YOLOv8 introduces many new features and improvements over previous versions, including:
2. Anchor-free architecture: YOLOv8 uses an anchor-free architecture, which makes it easier to train the model on different datasets.
3. Self-attention mechanism: YOLOv8 uses a self-attention mechanism in the head of the network, which helps to learn long-range dependencies between features better.
4. Adaptive training: YOLOv8 uses adaptive training to optimize the learning rate and balance the loss function during training, leading to better model performance.

Advanced data augmentation: YOLOv8 uses a number of advanced data augmentation techniques, such as MixUp and Mosaic, to improve the robustness of the model to different variations in the data. Other object detection algorithms inspired YOLO. The success and innovation of YOLO catalyzed a plethora of research in the realm of object detection. Many algorithms have taken inspiration from YOLO's design principles, attempting to replicate its success or address its limitations. Here's a look at some object detection algorithms that were influenced by or drew inspiration from YOLO:

Single Shot MultiBox Detector (SSD): SSD is another popular single-pass object detection system, which, similar to YOLO, aims at predicting category scores and box offsets for a fixed set of default bounding boxes. While SSD's design differs in many ways, it shares YOLO's philosophy of performing detection in one forward pass of the network.

CornerNet & CenterNet: Instead of predicting bounding boxes, these models detect objects by finding their corner or center points. While their approaches are distinct, the drive to simplify and streamline detection is reminiscent of YOLO's breakthroughs.

RefineDet: This single-pass detector refines object locations in a cascaded manner. By combining the benefits of two-stage detectors (like Faster R-

CNN) and single-pass detectors (like YOLO), it attempts to strike a balance between detection speed and accuracy. **M2Det:** This model focuses on multi-level feature extraction, drawing from the success of YOLO's multi-scale predictions. By extracting features at multiple scales, it aims to improve detection accuracy across differently-sized objects.

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CHAPTER-4

EXPERIMENTAL ANALYSIS AND DISCUSSION

8.1 Analysis:

First collection of image dataset from the nearby station with different objects ex:Tary,Helmet, Geometry, Bucket. Using the data augmentation increasing the dataset .By using the labelling technique bounding the text box for respective images using the yolo code giving the input images to the code along with the labelling and writing the yolo code and training and testing the data using the yolo code and predicting the output.



Fig 4.1.1 Tray as input for Data Augmentation Output:



Fig 4.1.2Tray as output for Data Augmentation



Fig 4.1.3 Tray as output for Data Augmentation



Fig 4.1.4 Bucket as input for data augmentation

Output:



Fig 4.1.5 Bucket as Output for data Augmentation



Fig 4.1.6 Bucket as Output for Data Augmentation

INPUTS FOR OBJECT LABELLING USING YOLO ALGORITHM



Fig 4.1.7 Tray as an input for Object Labelling



Fig 4.1.8 Helmet as an input for Object Labelling



Fig 4.1.9 Geometry box as an input for ObjectLabelling





Fig 4.1.8 Helmet as an input for Object Labelling



Fig 4.1.9 Geometry box as an input for ObjectLabelling



Fig 4.1.10 Bucket as an input for Object Lab Outputs of Labelling



Fig 4.1.11 Labelling of tray

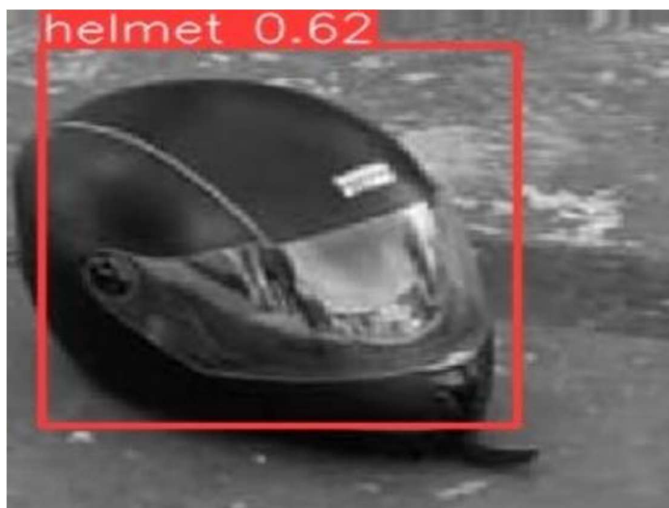


Fig 4.1.12 Labelling of Helmet



Fig 4.1.13 Labelling of Bucket

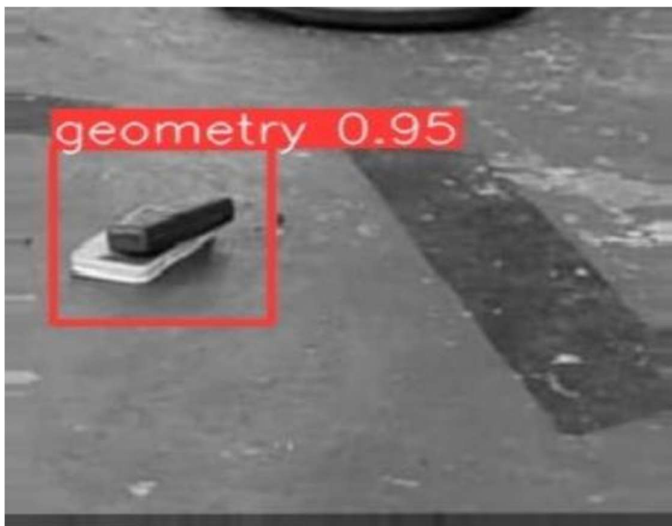


Fig 4.1.14 Labelling of Geometry box

8.2 Discussion:

The training process involved feeding the annotated images into the YOLO model, configuring the network architecture, and optimizing model parameters to minimize detection errors and maximize performance. During training, the YOLO model iteratively learned to detect and localize objects within the images based on the provided bounding box annotations.

The study embarked on a comprehensive exploration of object detection methodologies, centering on the utilization of the YOLO (You Only Look Once) algorithm, to analyze image datasets gathered from a nearby station.

The dataset encompassed a diverse array of objects, including tary, helmet, geometry items, and buckets. Leveraging advanced computer vision techniques, the research aimed to develop a robust object detection model capable of accurately identifying and localizing these objects within images.

The initial phase of the study involved the collection and preprocessing of the image dataset. Images were captured from the station's surroundings, ensuring variability in lighting conditions, object orientations, and backgrounds. To enhance dataset diversity and improve model generalization, data augmentation techniques were employed.

This involved applying transformations such as rotation, flipping, scaling, and color jittering to the original images, effectively expanding the dataset size while preserving object semantics.

Subsequently, the annotated dataset was prepared for training by labeling each image with bounding boxes delineating the objects of interest. The labeling process was facilitated using YOLO code, which allowed for efficient annotation of objects and generation of corresponding bounding box coordinates.

This step was crucial for providing ground truth annotations necessary for training and evaluating the object detection model. With the dataset labeled and preprocessed, the YOLO algorithm was implemented for training and testing the object detection model. The YOLO architecture is renowned for its speed and accuracy, making it an ideal choice for real-time object

detection tasks.

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CHAPTER-5

CONCLUSION AND FUTURE SCOPE

5.1 conclusion:

YOLO(You Only Look Once) algorithm for object detection and labeling with deep learning offers fast and precise identification of multiple objects in images. Its efficiency and straightforward approach make it a powerful tool for real-time applications like surveillance and automation.

Object detection, powered by deep learning techniques, has made significant strides in recent years. Researchers and practitioners have developed sophisticated models that can accurately locate and classify objects within images. These advancements have practical applications across various domains, including surveillance, autonomous driving, and medical imaging. The YOLOv8 object detection model presents a compelling solution in the realm of computer vision, offering significant advancements in accuracy, speed, and efficiency. Through its innovative architecture and training techniques, YOLOv8 has demonstrated remarkable performance across various datasets and real-world applications.

In conclusion, YOLOv8 represents a significant milestone in object detection technology, bridging the gap between accuracy and speed. Its ability to detect objects in real-time with high precision makes it an invaluable tool for numerous industries, including autonomous vehicles, surveillance, retail, and more. As research and development in computer vision continue to evolve, YOLOv8 serves as a testament to the ongoing pursuit of pushing the boundaries of what is possible in artificial intelligence and machine learning. YOLO-based object detection with labeling using deep learning could see advancements in real-time processing efficiency, expanded applications in edge computing, and increased accuracy through continuous model refinement, making it even more versatile and impactful in various domains.

5.2 Future scope:

In the realm of computer vision, YOLOv8 object detection stands as a beacon of innovation, offering a blend of accuracy and speed that promises transformative applications across numerous fields.

Looking ahead, the future scope for YOLOv8 is expansive and dynamic. Improving its accuracy through refined architectures and advanced training methodologies holds promise for achieving even greater precision in object detection tasks. Integration with semantic segmentation could enrich its capabilities, enabling deeper scene understanding and context-awareness. Moreover, enhancing YOLOv8's robustness to challenging conditions and optimizing it for real-time deployment on resource-constrained platforms will be essential for its practical utility. Additionally, advancements in multi-object tracking, continual learning, and domain-specific customization offer avenues for further expansion into specialized domains and applications. Ethical considerations, such as privacy and bias mitigation, will also play a critical role in shaping the responsible deployment of YOLOv8 in society. As research and development continue to push the boundaries of computer vision technology, YOLOv8 remains at the forefront, poised to redefine how we perceive and interact with the visual world.

Efficiency: Improving the efficiency of object detection models remains a priority. Faster inference times and reduced computational requirements are essential.

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APPENDIX - 1

Pip install (recommended)

```
!pip install ultralytics
```

```
## importing required libraries import os  
import shutil import random
```

```
!pip install tqdm --upgrade  
from tqdm.notebook import tqdm
```

```
from google.colab import drive drive.mount('/content/drive1')
```

```
train_path_img = "./yolo_data/images/train/" train_path_label =  
"./yolo_data/labels/train/" val_path_img = "./yolo_data/images/val/"  
val_path_label = "./yolo_data/labels/val/" test_path = "./yolo_data/test"
```

Split the dataset into train and test and creates the train.txt and test.tx with the
respective path of the images in each folder

'''

```
def train_test_split(path,neg_path=None, split = 0.2): print("----- PROCESS  
STARTED----- ")
```

```
# Pip install (recommended)
```

```
!pip install ultralytics
```

```
## importing required libraries import os
```

```
import shutil import random
```

```
!pip install tqdm --upgrade
```

```
from tqdm.notebook import tqdm
```

```
from google.colab import drive drive.mount('/content/drive1')
```

```
train_path_img = "./yolo_data/images/train/" train_path_label =
```

```
"./yolo_data/labels/train/" val_path_img = "./yolo_data/images/val/"
```

```
val_path_label = "./yolo_data/labels/val/" test_path = "./yolo_data/test"
```

Split the dataset into train and test and creates the train.txt and test.tx with the respective path of the images in each folder

```
'''
```

```
def train_test_split(path,neg_path=None, split = 0.2): print("----- PROCESS  
STARTED----- ")
```

```
files = list(set([name[:-4] for name in os.listdir(path)])) ## removing  
duplicate names i.e. counting only number of images
```

```

print (f'--- This folder has a total number of {len(files)} images---')
random.seed(42)
random.shuffle(files)
test_size = int(len(files) * split) train_size = len(files) - test_size

## creating required directories os.makedirs(train_path_img, exist_ok =
True)      os.makedirs(train_path_label,      exist_ok      =      True)
os.makedirs(val_path_img, exist_ok = True) os.makedirs(val_path_label,
exist_ok = True)

### -----copying images to train folder
for filex in tqdm(files[:train_size]):
if filex == 'classes': continue
shutil.copy2(path + filex + '.jpg',f'{train_path_img}/" + filex + '.jpg' )
shutil.copy2(path + filex + '.txt', f'{train_path_label}/" + filex + '.txt')
print(f'----- Training data created with 80% split
{len(files[:train_size])} images --- ")

if neg_path:
    neg_images = list(set([name[:-4] for name in os.listdir(neg_path)])) ##
removing duplicate names i.e. counting only number of images
for filex in tqdm(neg_images):
shutil.copy2(neg_path+filex+ ".jpg", f'{train_path_img}/" + filex + '.jpg')

print(f'----- Total {len(neg_images)} negative images added to the training
data-----")

```

```
print(f"----- TOTAL Training data created with {len(files[:train_size])} +  
-----  
len(neg_images)} images --- ")
```

```
### copytin images to validation folder for filex in tqdm(files[train_size:]):
```

```
## splitting the data into train-test and creating train.txt and test.txt files
```

```
# train_test_split('/content/drive/MyDrive/custom_notebooks/yolo_data/')
```

```
### for label_tag
```

```
train_test_split('/content/drive1/MyDrive/yolov8/data/') ### without  
negative images # train_test_split('./data','./negative_images/') ### if you  
want to feed negative images
```

```
!yolo task=detect mode=train model=yolov8s.pt  
data=/content/drive/MyDrive/yolov8/dataset.yaml epochs=10 imgsz=640  
batch=8 project=/content/drive1/MyDrive/yolov8/training_results  
name=found
```

```
!yolo task=detect mode=predict  
model=/content/drive/MyDrive/yolov8/training_results/found/weights/best.  
pt conf=0.55 source=/content/drive1/MyDrive/yolov8/test_images
```

```

if filex == 'classes': continue

# print("running")

shutil.copy2(path + filex + '.jpg', f'{val_path_img}/' + filex + '.jpg' )
shutil.copy2(path + filex + '.txt', f'{val_path_label}/' + filex + '.txt')


print(f'----- Testing data created with a total of {len(files[train_size:])}
images -----')


print("----- TASK COMPLETED- ")


## splitting the data into train-test and creating train.txt and test.txt files
# train_test_split('/content/drive/MyDrive/custom_notebooks/yolo_data/')


### for label_tag
train_test_split('/content/drive1/MyDrive/yolov8/data/')    ###    without
negative images # train_test_split('./data/', './negative_images/') ### if you
want to feed negative images


!yolo          task=detect          mode=train          model=yolov8s.pt
data=/content/drive/MyDrive/yolov8/dataset.yaml  epochs=10  imgsz=640
batch=8        project=/content/drive1/MyDrive/yolov8/training_results
name=found


!yolo          task=detect          mode=predict
model=/content/drive/MyDrive/yolov8/training_results/found/weights/best.
pt conf=0.55 source=/content/drive1/MyDrive/yolov8/test_images

```

```
!cp -r /content/runs/detect/predict /content/drive1/MyDrive/yolov8/output
```

```
from google.colab import drive drive.mount('/content/drive')
```